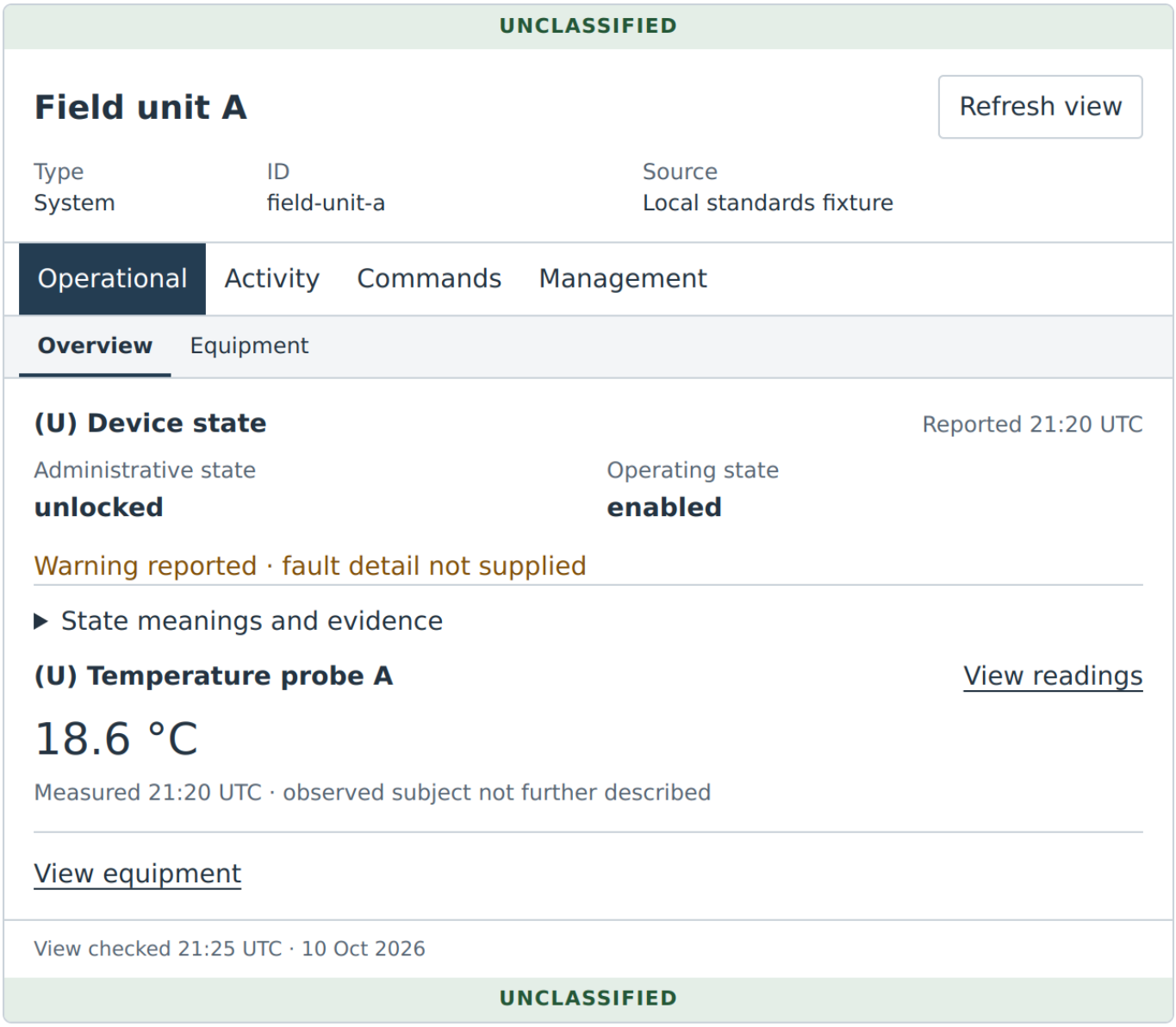
An overview with evidence behind its values

Standards fixture | All device names, readings and identifiers on this screen are synthetic.



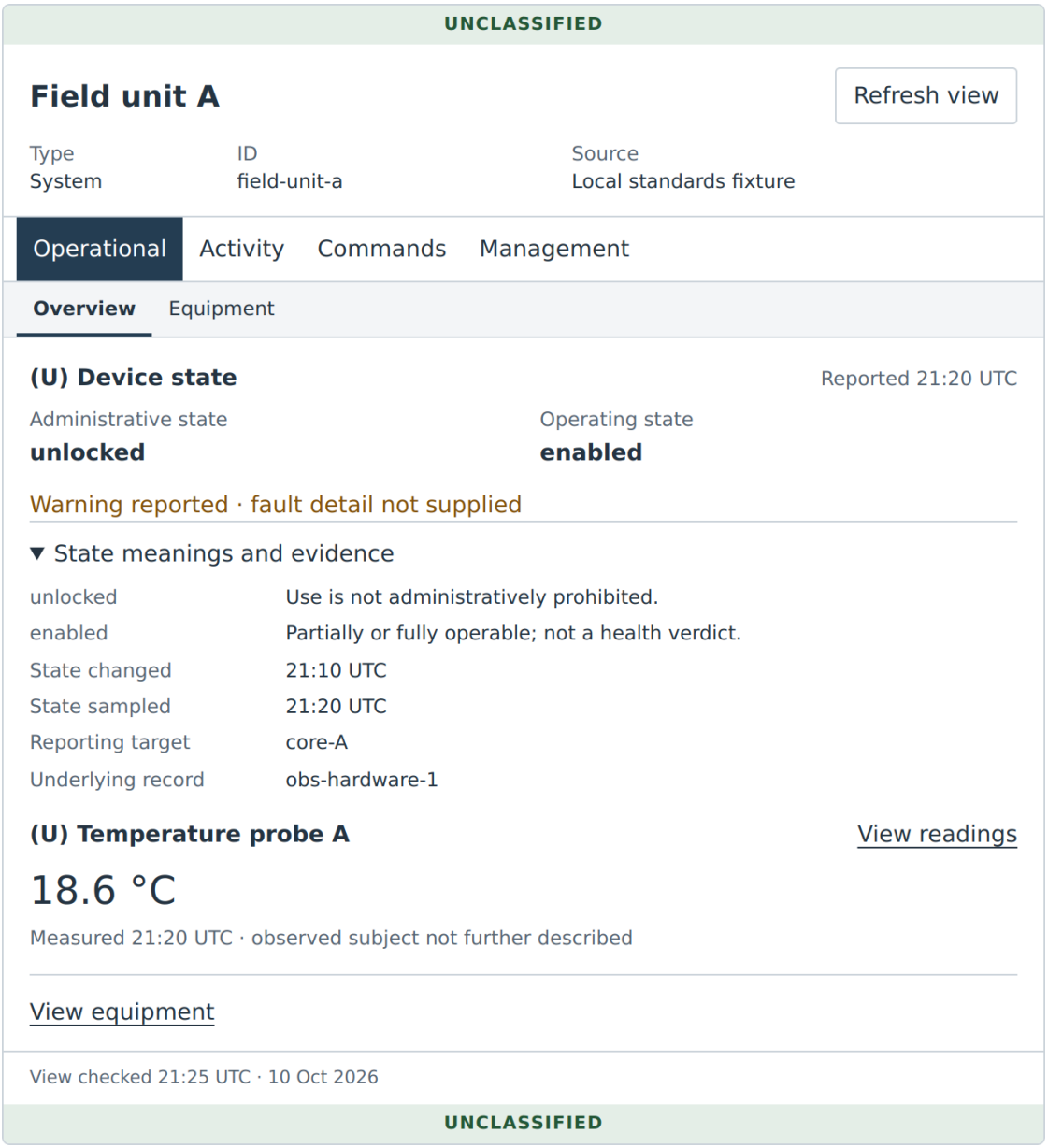
The visible change is the separation of administrative state, operating state and a reported warning. The hardware model gives these different meanings; it does not define a single universal green status. This example therefore allows an operable device and a warning to coexist. It keeps the source codes visible, with an explanation one click away. [S1, S2]

This is a focused test of information we can recognize, not a new decision that every system must open with these fields. Your manual-first operational priorities still govern the eventual default. A system with verified emplacement information may prioritize that instead. The compact frame, fixed identity and two tab levels remain; a known reading and equipment link give access to the particular system without making a dashboard.

Proposed binding: a typed hardware-state observation supplies the state block; a related temperature datastream supplies the reading. An integration must establish both subjects and associations. The fixture declares them; the manuals do not. The example markings demonstrate placement only. [S8, S9]

Expansion answers what the status actually means

Operational / Overview | Same card, with one section expanded.



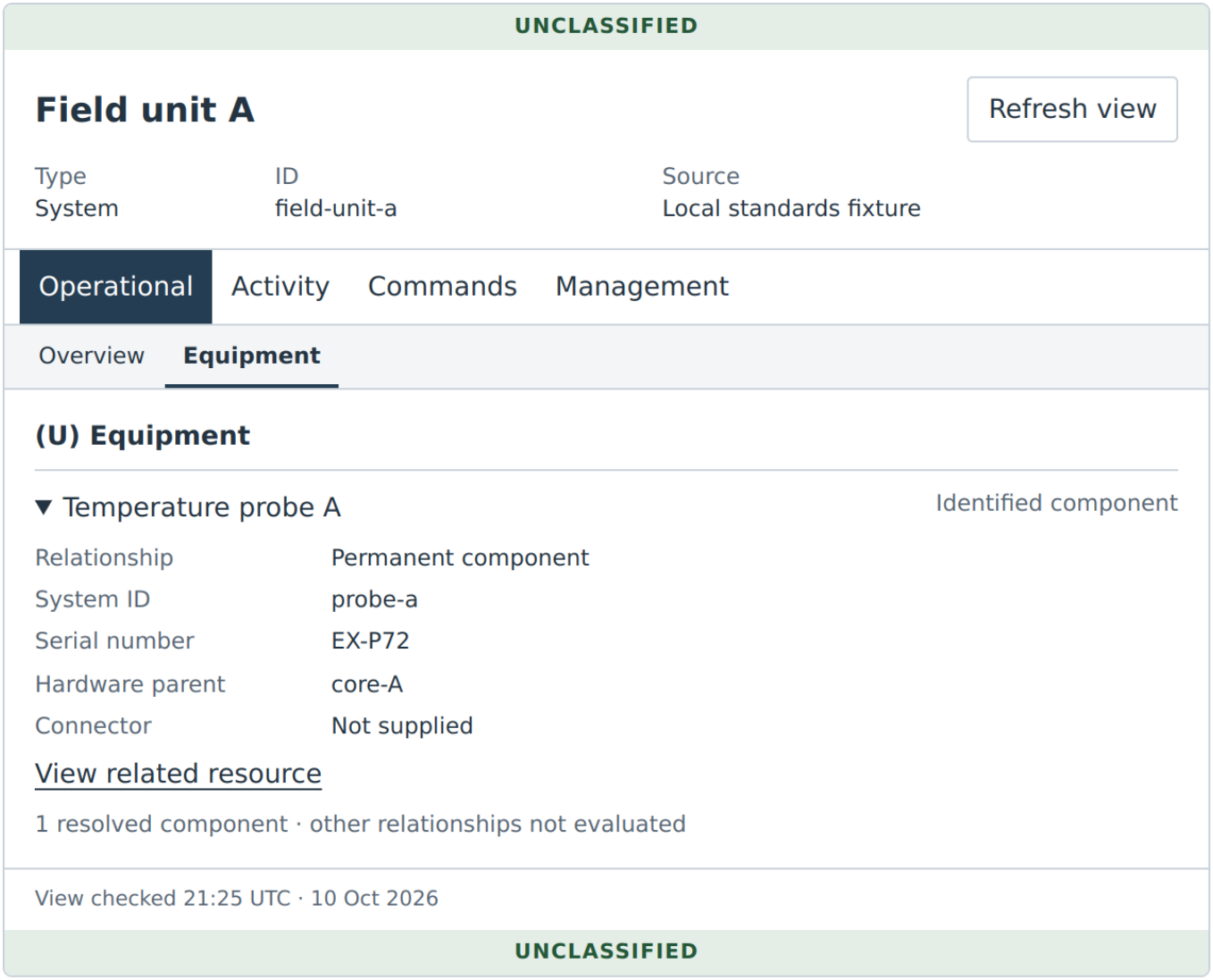
The expanded view is where the distinction becomes understandable. The state-change time, the time a state was sampled, and the time this view was checked answer different questions. A later refresh must not make an old observation look new. The RFC gives us a state-change property; the observation and the client supply the other time roles. [S1, S8]

I used a short disclosure row because most users need the values first, while someone making a consequential decision needs to inspect their basis. Expansion stays beside the value that prompted the question. The warning does not invent a failed part or diagnostic cause, because the example has only the warning flag. Its absence would likewise require careful treatment rather than automatic reassurance.

Proposed binding: state-last-changed remains separate evidence; it is not copied into every observation timestamp. This fixture assumes the adapter sampled the displayed state at 21:20. That sampling assertion must be recorded by the real integration. “Refresh view” rereads local example data and never requests a new device report.

Equipment becomes navigable when identity is known

Operational / Equipment | Synthetic, individually identified component.



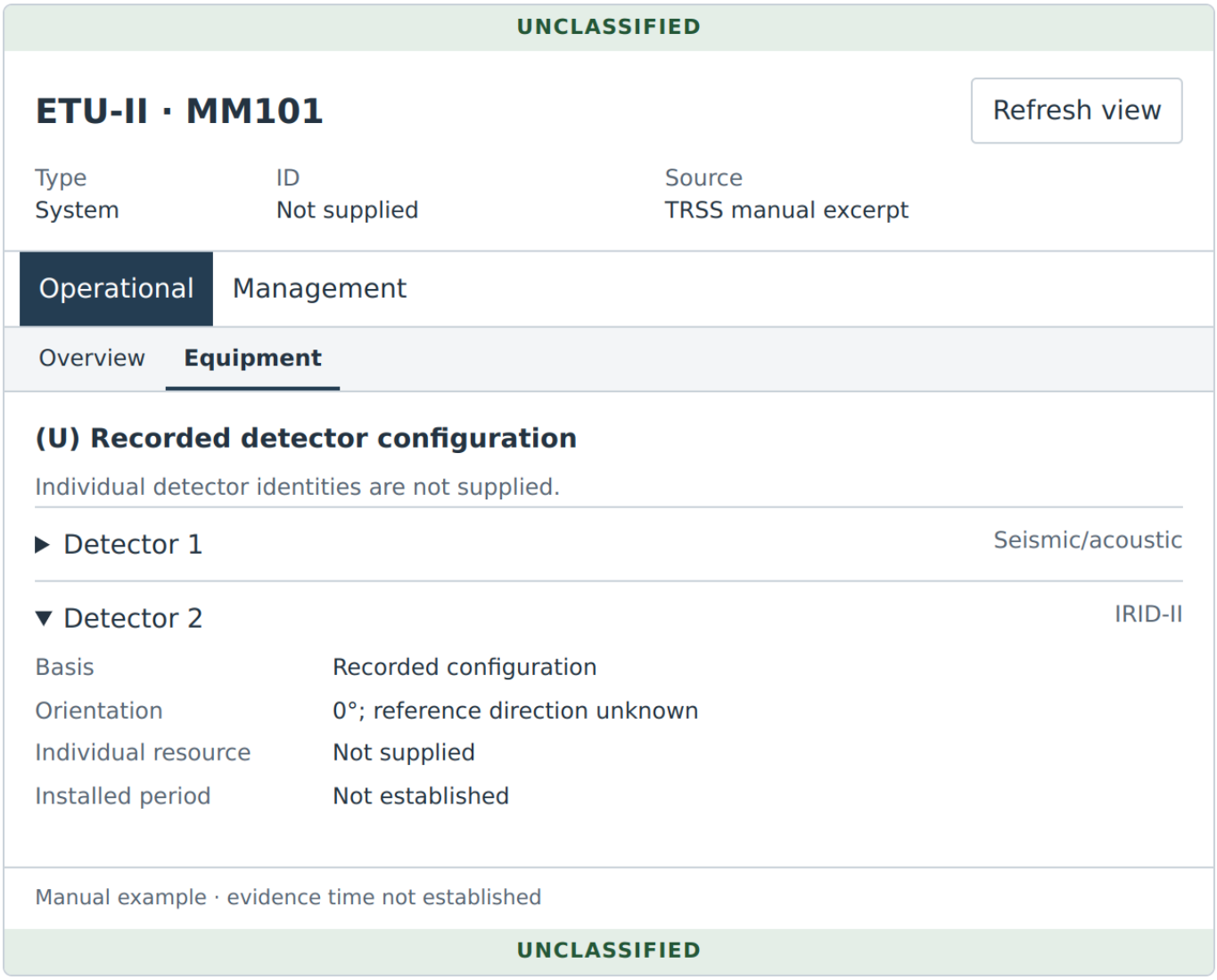
This row describes an actual modeled instance: a particular probe with an identifier and serial number. Its expansion can therefore offer a related-resource preview. The equipment list is a useful common destination even though different integrations may supply different kinds of evidence. RFC hardware containment contributes a parent relationship, while OPC Devices reinforces the separation between supported types and configured instances. [S1, S4]

Neither of those facts alone decides that a component is permanently part of a CSAPI System. For this exercise, a separate integration assertion establishes that relationship. A swappable attachment may instead need a time-qualified deployment association. The screen keeps the relationship visible and does not claim to know a connector simply because it knows a parent. This is how the common presentation can grow richer without changing the meaning of a sparse record.

Proposed binding: resolve the hardware component key through a source-qualified resource lookup, then use the verified CSAPI relationship. The preview demonstrates navigation only. No production endpoint or automatic RFC-to-CSAPI containment translation is implemented here. [S7]

The same destination with less evidence

Operational / Equipment | TRSS manual values carried forward from the worked comparison.



The TRSS example remains deliberately different. Detector 1 and Detector 2 identify configuration positions, and the inspected manual supplies selected detector types. It does not identify two separately addressable sensor instances. Treating the rows like the identified probe on the preceding page would manufacture information. The new research is useful here because it disciplines what an equipment row is allowed to mean. [P1, P2, S4]

The row pattern remains consistent: label, concise value, then expansion. What expands depends on evidence. Zero degrees survives as a real value, but the unknown reference direction prevents a confident map bearing. No View Details link appears for a detector whose resource identity is unknown. The absence of a deployment record also remains unresolved, rather than proving that the detector is permanently attached or absent.

Proposed binding already examined: SensorML detector parameters using SWE Category and Quantity. The vendor-specific parameter definitions remain a reviewed adapter/profile responsibility. This pass does not assign RFC hardware identities to the manual screenshot. [P1, S6, S9]

One temperature view, three distinct meanings

Activity / Charts | OMA Temperature Object 3303 semantics; synthetic values.



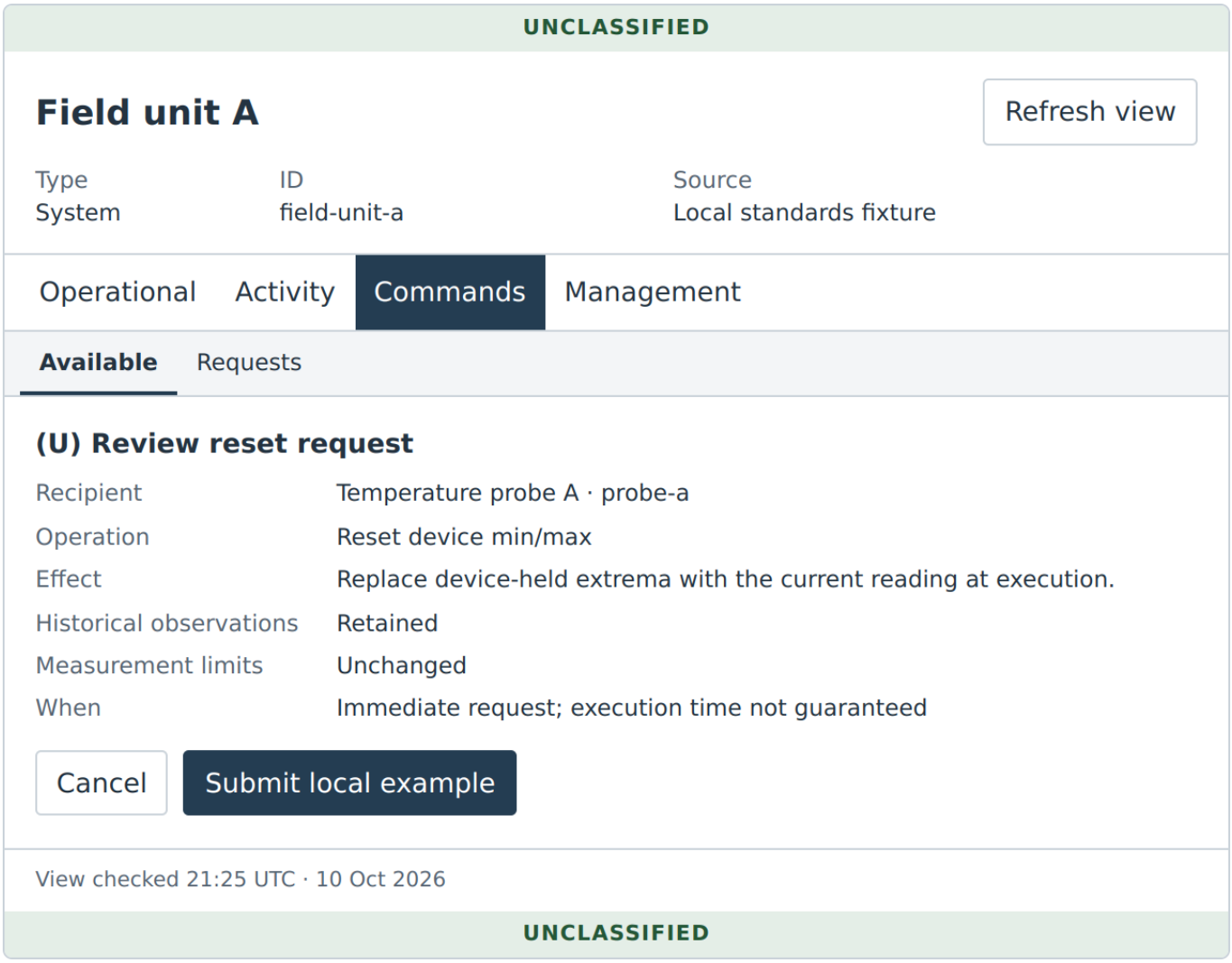
This is where the newer domain research visibly changes the earlier generic chart. The current reading is 18.6 degrees Celsius. The chart shows three loaded observations. Below it, the device remembers different extrema, while a separate disclosure shows what it can measure. OMA defines these as separate resources; the client must not merge them into a single ambiguous range. [S5]

The resulting UI is intentionally quiet. The reading gets emphasis, the trend provides quick orientation, and capability information stays compact. Device extrema have their own accumulation period, so they need not equal the extrema of the visible chart. The -40 to 85 range is not painted as a safe band: measuring capability does not establish operational alarm thresholds. Joining the three points illustrates a trend and does not assert measurements between them.

Proposed binding: 5700/5701 feed a SWE Quantity; 5601/5602 feed distinct result fields; 5603/5604 feed a SensorML capability range. Resource 5518 supplies measurement time when present. These are proposed MERLIN bindings, not a published OMA-CSAPI crosswalk. [S5, S6, S8, S9]

A domain action, available where it is useful

Commands / Available | Reached from the reading view; local review only.



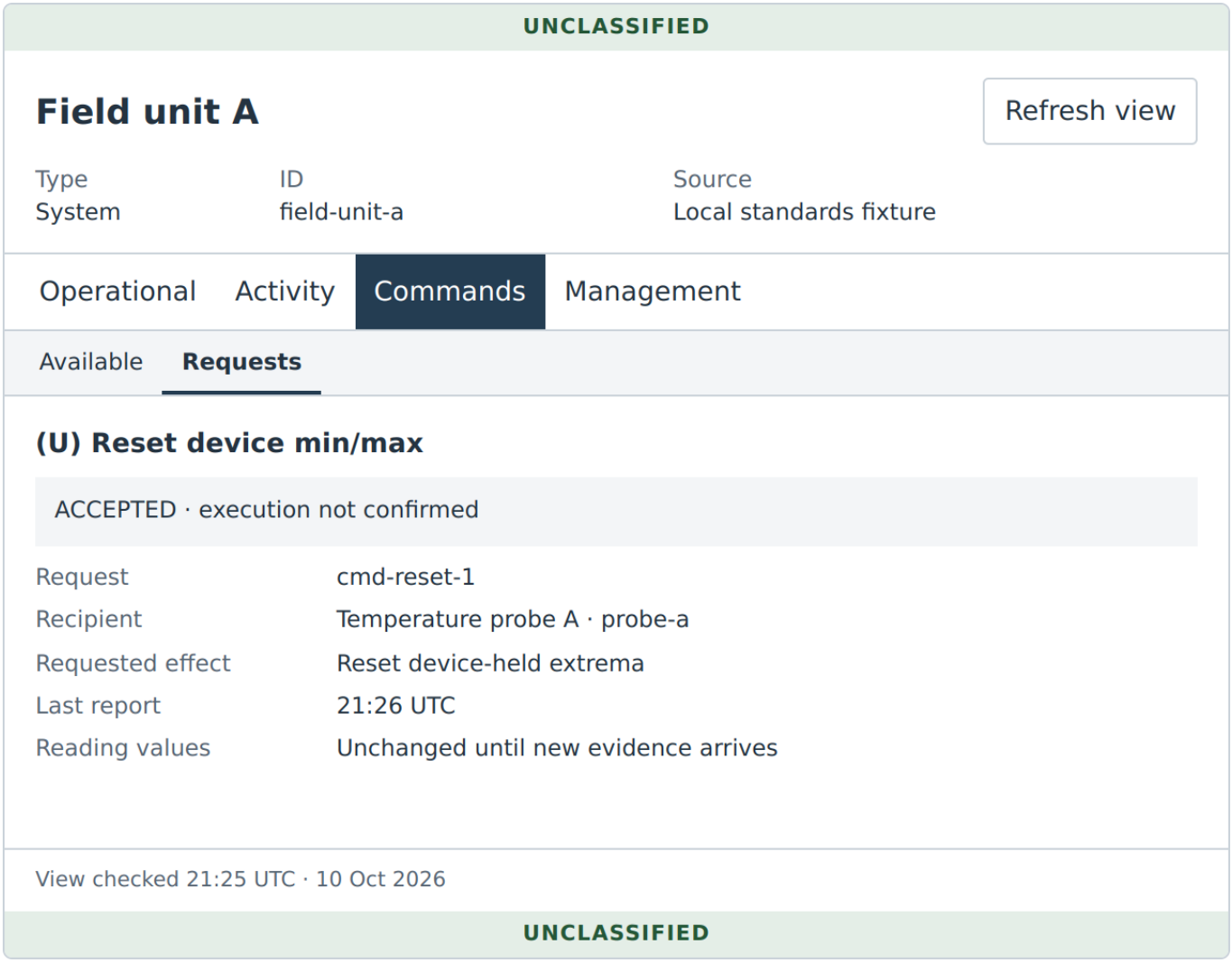
The reset action gives the new research a concrete interaction consequence. OMA resource 5605 concerns the device-held extrema. That is why the button sits near those values and why the review names the probe as its recipient. It is not an edit to the temperature reading. Historical observations remain intact in our proposed workflow. [S5]

Selecting the shortcut moves into the shared Commands workflow. We do not need a separate execution mechanism hidden inside every measurement widget. The review explains scope and effect before submission, including what remains unchanged. This also illustrates the useful OPC Functional Group precedent: one underlying operation can be relevant in more than one task grouping. The particular tab placement is our design decision. [S3]

Proposed binding: a verified ControlStream exposes a command whose adapter invokes 3303/{instance}/5605. An advertised OMA operation alone does not create that route or authorize the user. The local button says Submit local example so nobody mistakes this wireframe for a connected control. [S8]

An accepted request has not yet changed the sensor

Commands / Requests | Synthetic acknowledgment, with no execution report.



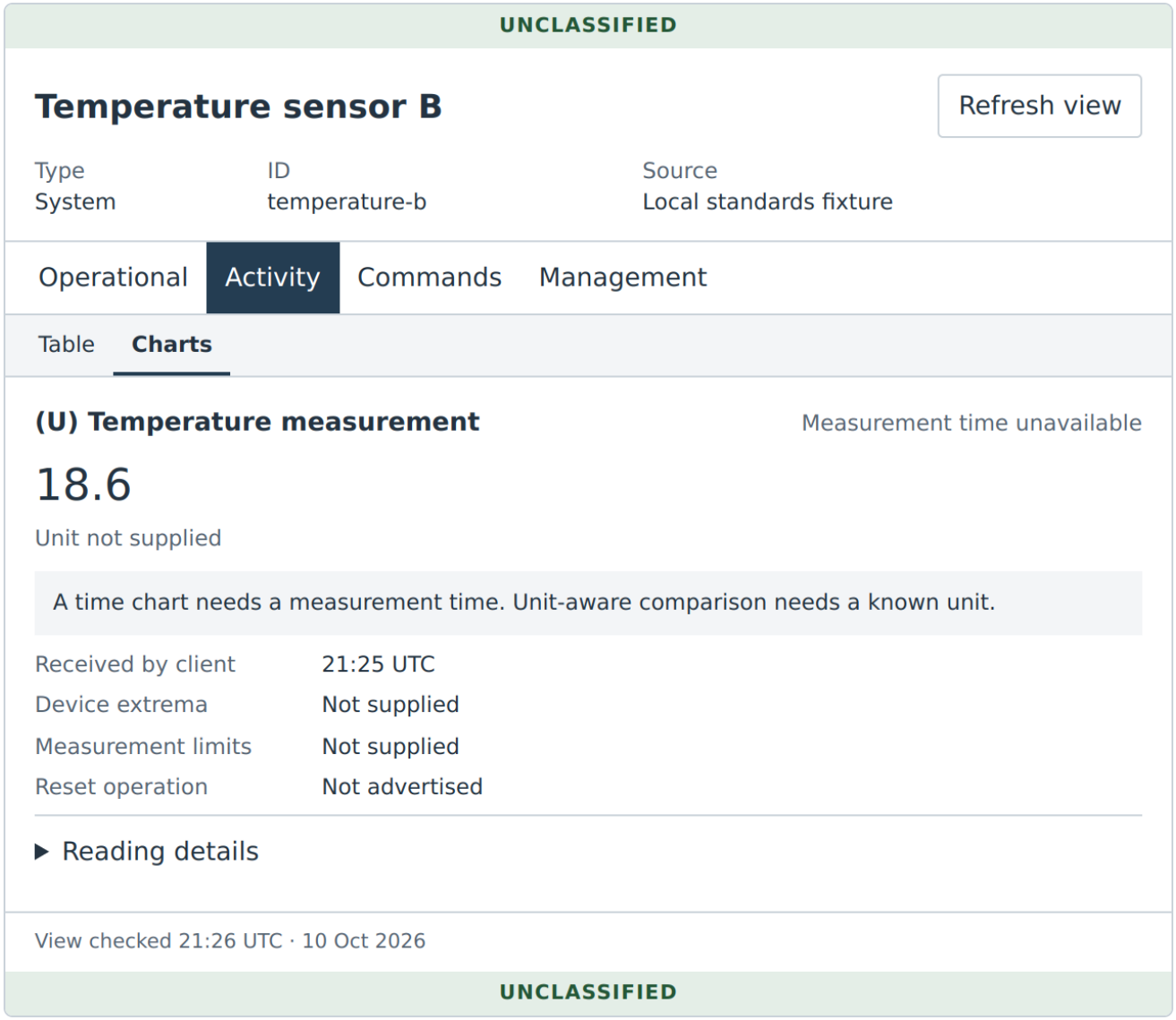
After submission, the request receives an identity and an ACCEPTED state. CSAPI distinguishes acceptance from later execution outcomes. The screen therefore leaves the existing extrema alone. Optimistically replacing them with the latest displayed temperature would be misleading: the current value at actual execution could be different, and the command could fail. [S8]

This is a presentation decision with a practical consequence. Users should be able to move between a request and the information it is intended to affect without losing the distinction between intent and evidence. A real implementation would follow status reports and correlated results, then show newly returned device values. An accepted request must remain visible even if the user leaves the Commands category.

The example exercises review, cancel, local submit, request identity and refresh. It does not exercise transport, permissions, retry/idempotency, cancellation after submission, failure recovery or hardware execution. Those are acceptance work for the actual command integration; a working button in this packet does not establish them.

Less information should produce an honest smaller claim

Activity / Charts | Synthetic source omits optional units, timestamp and extrema.



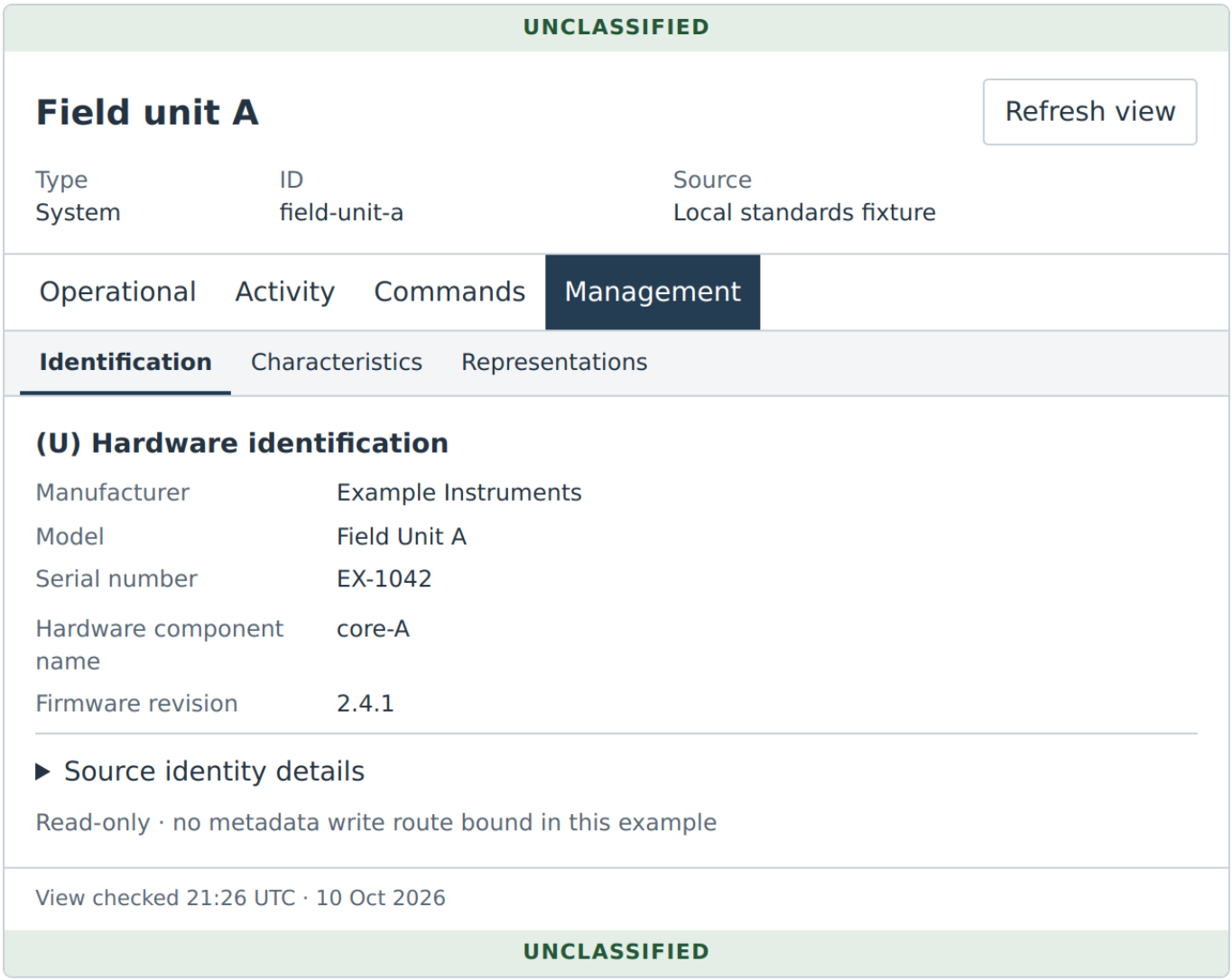
The numeric reading survives, but the temperature unit and measurement time do not appear by assumption. Object 3303 permits those supporting resources to be optional. A source can therefore follow the object model and still fail to supply enough for our preferred chart. This is an important limit of the media analogy: knowing the object family does not guarantee every useful display input. [S5]

Keeping the user at the same destination makes the missing information understandable. We explain why a chart cannot be drawn, keep the supplied value inspectable, and omit the unavailable reset action. “Not supplied” does not mean zero, safe, broken or access denied. Production retrieval must preserve those different outcomes. The visible receive time is deliberately not substituted for measurement time.

This sparse example is a source-level fallback. With this binding, it is not promoted into a timed CSAPI observation until a defensible phenomenon time is available. A documented acquisition-time meaning could be used by an integration, but silently inventing measurement time is not acceptable. A known unit from another authoritative descriptor could likewise complete the mapping.

Management explains which hardware this is

Management / Identification | Read-first; synthetic metadata.



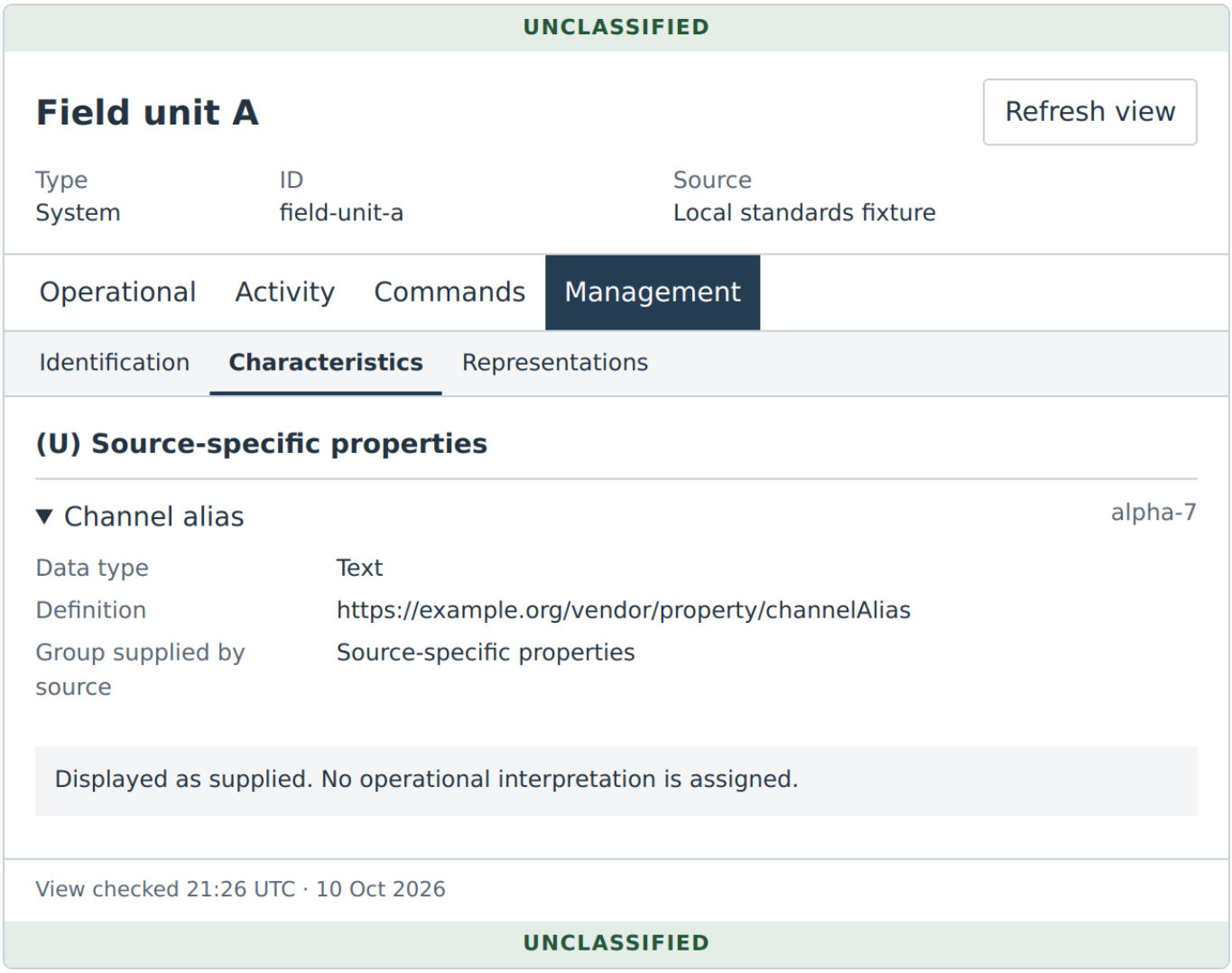
Manufacturer, model, serial number and firmware revision answer different identity questions. The additional hardware vocabulary helps us preserve those roles instead of displaying a flat collection of unexplained identifiers. The fixed header still carries resource Type, ID and service Source; the manufacturer must not replace the source server. [S1, S7]

I kept these stable descriptive details under Management so they remain accessible without dominating operational use. A useful identifier can also appear near an equipment row without becoming a second independently maintained value. The page is read-first. Editing should be enabled per property and route: an editable alias does not imply that a manufacturer-assigned serial number or reported firmware version should have an edit box.

Proposed binding: selected hardware identity leaves populate defined SensorML identifiers; firmware is a typed characteristic. The draft definition URIs used in the fixture are placeholders. This establishes carrier shape, not an approved ontology or a writable server transaction. [S6, S9]

Unfamiliar information remains accessible

Management / Characteristics | A known data type with an unresolved operational meaning.



Channel alias is readable because its structural type is text. Its source grouping and definition remain inspectable. Nothing in that structure proves that alpha-7 is a radio channel, a mission assignment or a communications setting. This is the limit that the client must respect even after we add domain standards: recognized structure is not the same as recognized meaning. [S6, S9]

This view preserves the strongest part of the Integrator approach while allowing a more useful front door. The complete descriptive record remains available, and reviewed mappings selectively elevate information into operational views. We can add a specialized presentation for a newly understood concept without forcing every other vendor to supply it or erasing its original group. The fallback is therefore a deliberate part of the design, not a failed integration screen.

The proposed client selects a mapping by definition and supported version/profile, or by an explicitly reviewed legacy adapter. Labels alone must never select a trusted operational interpretation. This fixture demonstrates the fallback result; it is not a general-purpose implementation of a mapping registry. [P1]

What we are actually reusing

A small semantic layer between source information and presentation; not a replacement for CSAPI.

The practical architecture

A source adapter or server-side profile interprets the incoming device model. It emits OGC resources with explicit meanings, typed values, units, subjects and time. The client recognizes supported definitions and applies a reusable view. These responsibilities could be colocated, but the client should not need a separate native protocol stack for every standard referenced in the research.

What this changes about our approach

We stop choosing a visual solely because a field is called status, battery or range. We choose it because the integration has established a particular meaning. Once that meaning is known, different vendor paths can use the same display. The result is repeatable specialization inside the common card. It is closer to the image/media analogy you proposed, but with a crucial additional step: domain meaning often needs an explicit crosswalk.

What remains a MERLIN decision

Operational, Activity, Commands and Management are still proposed task groupings. The standards do not impose these tab names, the default order, our 640 CSS-pixel review width, or the choice to elevate a reading. The manual evidence and your access-first requirement govern those choices. OPC Functional Groups provide a precedent for task grouping and referencing, not a normative MERLIN layout. [S3]

Research family	Disposition in this increment
RFC 8348 / RFC 4268	Selected semantics exercised: hardware identity, containment and distinct states. No YANG runtime or protocol conformance claim.
OPC UA Devices	Organizing precedent and type/instance distinction. No OPC server, NodeSet ingestion or OPC conformance claim.
OMA Temperature 3303 v1.1	Selected resources exercised through an illustrative OGC crosswalk. No full-object or LwM2M protocol implementation.
AAS / MIMOSA	Deferred candidates for broader asset/lifecycle needs. No new template, runtime or license dependency adopted.
DISR and openness	Retain the earlier version-specific assessment as a selection gate. This packet neither changes status nor treats a listing as a field mapping.

Trace the hardware information

Exact source selectors and proposed targets; these are integration proposals, not mappings mandated by either SDO.

In the source column below, H means /ietf-hardware:hardware/component[name="core-A"]. Bracketed selectors describe named entries; they are not literal JSON Pointers. The fixture pins ietf-hardware to its 2018-03-13 module revision. Definition URIs beginning https://example.org/merlin-draft/device/0.1/ identify this exercise only. They must be replaced by reviewed, versioned semantic bindings before deployment.

Source concept / selector	Proposed OGC carrier	Presentation rule
H/mfg-name; H/model-name; H/serial-num [S1]	SensorML identifiers[] entries, each with its own definition, label and value. [S6]	Manufacturer, model and serial remain distinct; none substitutes for service Source or API ID.
H/firmware-rev [S1]	SensorML characteristics[].characteristics[]: Text named firmwareRevision.	Readable metadata; no inferred edit route.
H/state/admin-state; oper-state [S1, S2]	DataStream schema resultSchema: DataRecord with separate Category fields administrativeState and operatingState. [S8, S9]	Display separate codes with explanations. Preserve unknown or unsupported codes without inventing a health score.
H/state/alarm-state [S1, S2]	A DataArray of Category flags in the state result; a proposed adaptation of the source bit set.	A set of flags, not a single exclusive status. Missing, explicit unknown and an empty known set remain distinct.
H/state/state-last-changed [S1]	Separate defined time evidence; observation phenomenonTime is the sampled state time in this fixture.	Do not substitute change time or fetch time for sampling time. The compact example retains source change time in disclosure.
component[name="probe-A"]/parent = core-A [S1]	Explicit lookup to resource IDs, then a reviewed System composition or Deployment association. [S7]	No inference of permanence, connector or emplacement from a parent key alone.
TRSS detector choice [P1, P2]	SensorML parameters[] Category; orientation Quantity. [S6, S9]	Configured type does not create an identified instance or prove a physical connection.

Worked example: the source state is unlocked / enabled / warning. The proposed observation result carries three separate fields. The renderer chooses two labeled values and one warning statement, with no derived healthy flag. The OGC result structure passes a JSON-schema check; source semantics and the adapter assertion still require review.

Not yet bound: no actual TRSS or OmniSense API payload, source component-to-resource lookup, live hardware report, or production definition registry was supplied. The table makes these gaps explicit rather than hiding them behind an attractive screen.

Trace the temperature information

OMA object URN urn:oma:lwm2m:ext:3303:1.1; accessed registry definition, 10 Oct 2026.

The fixture uses object instance 0 for the probe. The CSAPI datastream and controlstream identifiers are separately assigned; an OMA object instance number is not automatically a CSAPI resource ID. A binding record must preserve that association, the source model revision, unit rules and the observed subject. [S5, S7, S8]

OMA resource [S5]	Proposed representation	Visible consequence
3303/0/5700 with 5701	Observation result.temperature; SWE Quantity with a reviewed definition and uom.code=Cel.	18.6 °C; no Celsius default when the unit is unknown.
3303/0/5518	Convert its time encoding to observation phenomenonTime only when it refers to that reading.	Measured 21:20 UTC; independent of client refresh and result availability.
3303/0/5601 and 5602	Distinct result.minMeasured and result.maxMeasured quantities; preserve their accumulation basis.	Device extrema stay separate from min/max computed over the visible history.
3303/0/5603 and 5604	SensorML capabilities[].capabilities[]: QuantityRange named temperatureLimits, value [-40,85].	Measurement limits remain capability information. No alarm thresholds are derived.
3303/0/5605	Verified ControlStream; proposed parametersSchema contains operation=resetMeasuredExtrema. Adapter executes the bound device operation.	One reset workflow, reachable from readings and Commands. Permission and route are independent prerequisites.
Only 5700 is provided	Preserve the source value. This draft binding cannot emit a fully timed, unit-qualified observation without additional evidence.	Readable fallback; no fabricated chart, unit, timestamp or reset operation.

The operation field is a MERLIN adapter proposal, not an OMA input parameter. A dedicated controlstream could use another valid command schema. The tested example checks its SWE structure only; it does not validate the complete command exchange. Similarly, QuantityRange is a carrier choice for paired bounds, not a rule that every capability must be flattened into a range.

The registry is mutable: the exercise records object version 1.1, but production must pin the registry revision or checksum as well. Other optional resources, including quality information and fractional time, are outside this selected mapping. Their omission here is not a claim of full 3303 coverage.

What can enter the PBI now

Candidate requirements and bounded evidence; owner review is still needed for placement and labels.

DV-01 Meaning before promotion

Promote a field only through a supported definition/profile or a reviewed adapter for an identified source schema version. Preserve original values and unresolved fields.

DV-02 Separate state meanings

Do not collapse administrative permission, operability, alarms, usage or vendor planning status into one healthy/online flag. Show only established meanings.

DV-03 Evidence-aware equipment

Distinguish a supported type, configured choice, reported label and identified resource. Enable linked details only when identity and association are resolved.

DV-04 Domain-aware quantities

Keep reading, measured extrema, measurement limits and alarm thresholds distinct. Require known units and time semantics for corresponding visualizations.

DV-05 Honest actions

Bind each action to an actual permitted route and recipient. Preserve request identity and distinguish acceptance from execution. Keep source values until new evidence arrives.

DV-06 Compact, readable fallback

Retain the module frame and local expansion pattern. Unknown or incomplete information stays accessible without guessed labels, units or relationships.

What was actually verified

Five official OGC JSON-schema checks passed for the sample carriers: System identity/characteristics, temperature capability range, temperature result structure, hardware state structure and command parameters. Five additional local semantic assertions checked key distinctions. The interactive example passed 78 local behavior/layout checks, including 320, 640 and 736 CSS-pixel widths; no JavaScript runtime errors were recorded. These counts are separate kinds of evidence, not an interoperability score.

Next acceptance step

Review one view at a time, then bind one actual source descriptor and payload to the proposed record. Test unknown codes, missing units/time, conflicting identity, permissions and command failure. Do not start another broad framework survey to close these narrow gaps. Map, Symbol, Power, deployments, contacts, imagery and other resource types remain in scope. Earlier Activity feed/live and editing work is not removed by this focused exercise.

Direct references and their role

Primary online sources checked 10 October 2026. UI choices and crosswalks are explicitly proposed.

S1 Bierman, Andy, Martin Bjorklund, Jie Dong, and Dan Romascanu. 2018. A YANG Data Model for Hardware Management. RFC 8348. Sections 3 and 7.1; ietf-hardware revision 2018-03-13.

https://www.rfc-editor.org/rfc/rfc8348.html

Hardware fields, containment and time/state carrier. These are semantic inputs, not an approved CSAPI crosswalk.

S2 Chisholm, Sharon, and Dan Romascanu. 2005. Entity State MIB. RFC 4268. Textual conventions for administrative, operational and alarm states.

https://www.rfc-editor.org/rfc/rfc4268.html

The underlying state distinctions help explain why one health indicator would lose meaning.

S3 OPC Foundation. OPC Unified Architecture Part 100: Devices. Section 4.4.1, FunctionalGroup model. Online edition accessed 10 October 2026.

https://reference.opcfoundation.org/specs/OPC-10000-100/4.4.1

Supports application-oriented grouping and references from multiple groups. Does not prescribe our tabs.

S4 OPC Foundation. OPC Unified Architecture Part 100: Devices. Section 9.2.1, Configurable components: general pattern. Accessed 10 October 2026.

https://reference.opcfoundation.org/specs/OPC-10000-100/9.2.1

Separates supported types from configured instances; no OPC runtime adoption is claimed.

S5 Open Mobile Alliance. LwM2M Object Registry, Temperature object 3303, object version 1.1. prod branch accessed 10 October 2026.

https://raw.githubusercontent.com/OpenMobileAlliance/lwm2m-registry/prod/3303.xml

Resources 5700, 5701, 5601-5605 and 5518 drive the worked temperature example. Pin a registry revision before implementation.

S6 Open Geospatial Consortium. OGC SensorML Encoding Standard, version 3.0. OGC 23-000. Sections 8.2 and 9.1.4.

https://docs.ogc.org/is/23-000/23-000.html

Identifiers, characteristics and capabilities provide proposed descriptive carriers.

Direct references and their role

Primary online sources checked 10 October 2026. UI choices and crosswalks are explicitly proposed.

S7 Open Geospatial Consortium. OGC API - Connected Systems Part 1: Feature Resources, version 1.0. OGC 23-001. Sections 8-11.

https://docs.ogc.org/is/23-001/23-001.html

Resource identity, Systems and composition/deployment relationships.

S8 Open Geospatial Consortium. OGC API - Connected Systems Part 2: Dynamic Data, version 1.0. OGC 23-002. Sections 9-10 and 16.

https://docs.ogc.org/is/23-002/23-002.html

Datastream schemas, observations, controlstreams, command parameters and command status.

S9 Open Geospatial Consortium. OGC SWE Common Data Model Encoding Standard, version 3.0. OGC 24-014. Component types and JSON encoding.

https://docs.ogc.org/is/24-014/24-014.html

Typed semantic values. Official schemas also retrieved from schemas.opengis.net/sweCommon/3.0/json/ and sensorML/3.0/json/.

P1 MERLIN Device Presentation Worked Comparison, 10 October 2026, current file version 1. Sections 1-10 and source register.

Existing six-subject comparison and equipment crosswalk. This packet now directly cites the additional standards instead of relying on this supplement alone.

P2 Sentinel 1.0.7.2 Software Users Manual, N65236-TRSS-SUM-0005-2.00. Supplied TRSS Details View Research deck, slides 10-11; reader pages 175-176.

TRSS values are carried forward through the directly inspected manual evidence recorded in P1. The screenshots do not supply API payloads.

P3 MERLIN Semantic Presentation and Interaction Guide v0.5, current file version 2; Resource Details Research Paper v0.15, current file version 20.

Existing project continuity: access-first presentation, identity, source meaning, progressive expansion and resource-specific behavior. No changes to these baseline documents are made by this packet.

Continuity note: this revision replaces the earlier wireframe review packet as the focused demonstration of the newer standards research. Prior mockups remain design history; missing views here have not been removed from the PBI. The master research paper, requirements baseline, semantics guide and separate Power Source workstream are not silently amended.

A common organization can carry both systems

Start here for this iteration. Earlier pages 1-16 remain the standards demonstrations; the comparison below adds manual evidence.

We have made progress on the original organizing problem. The manuals can fit one predictable set of destinations without making their fields falsely identical. The useful invariant is the user’s question: where is this system, how is it equipped, how does it communicate, what has it reported, and what can I do with it? The exact default fields and tab labels remain proposals.

Proposed category	Question it answers	Worked treatment
Operational	What do I need to access and understand this particular system as employed?	Overview; Equipment; Communications; Condition. Placement stays prominent when supplied.
Activity	What happened, and what information has this system produced or received?	Shared table with typed records; charts where the data supports them. History is not all one resource type.
Commands	What can I request, and what happened to my request?	Model-specific operations use one review/status workflow. No route or permission is assumed.
Management	What is the full record, and how may it be maintained?	Readable record and configuration; edit only through an established route. Source representations stay accessible.

Where the newer research matters

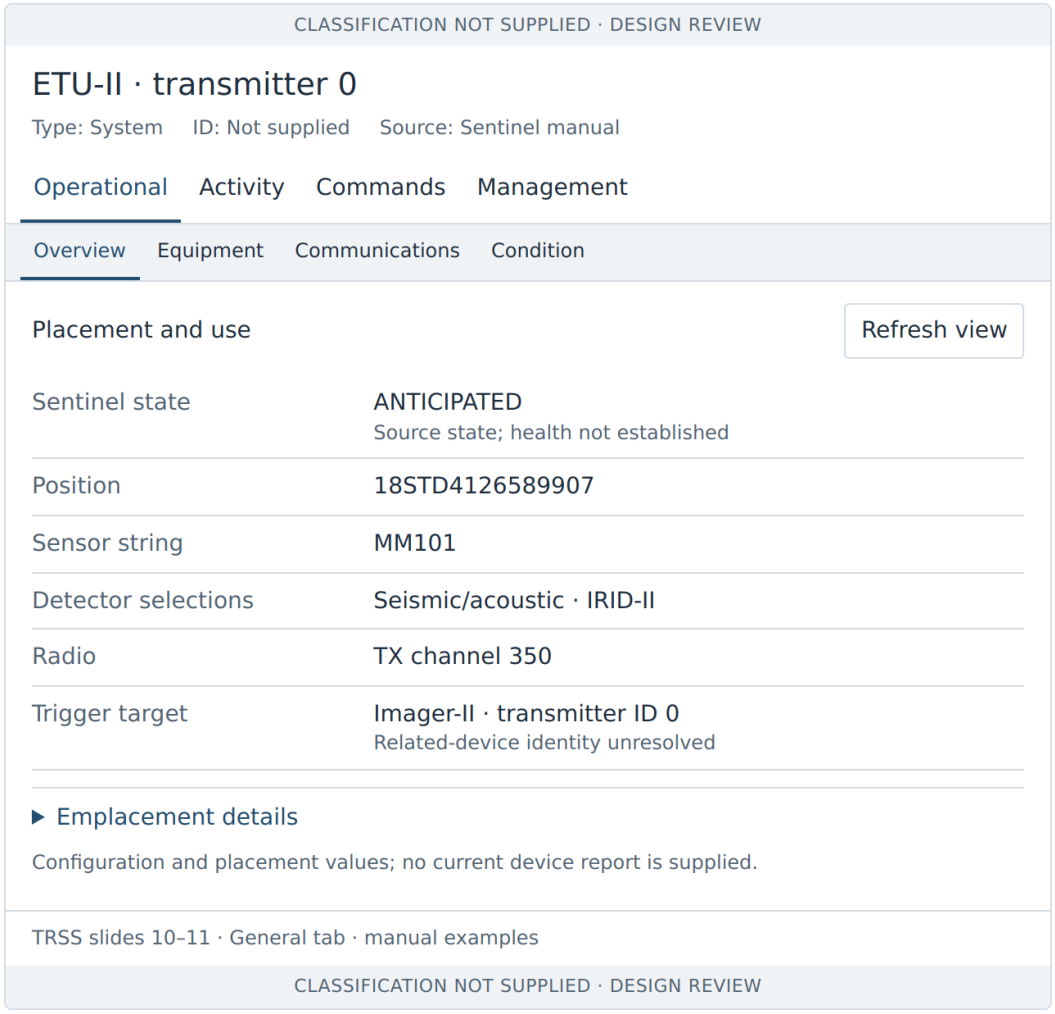
OPC UA Devices offers functional groups with recognized purposes, including Operational, Configuration, Diagnostics and Status. We use that as precedent for purpose-based organization and reuse of underlying information. These are not mandated MERLIN tabs, and the OPC meaning of Operational does not itself define military emplacement or assignment. Our operational priorities still come from your manuals and earlier MASBUS work. [D1]

The 40-subject ledger distinguishes shared concepts/carriers, vendor-specific bindings, and unresolved interpretations. It accounts for the selected excerpts; it does not claim complete manual, SensorML, CSAPI or production-device coverage. No actual API descriptors or payloads were supplied. The UI is therefore a reviewable design, not a validated integration.

Read pages 18-19 first to judge the front door. Pages 20-25 explain equipment, condition, configuration and commands. The remaining pages provide the trace and acceptance work. Map, Symbol, contacts, media, deployment navigation, feeds and live mode remain in scope; their final placement is not settled by this slice.

TRSS: start with the employed system

Operational / Overview | ETU-II manual examples, not live device data. [T10-12]



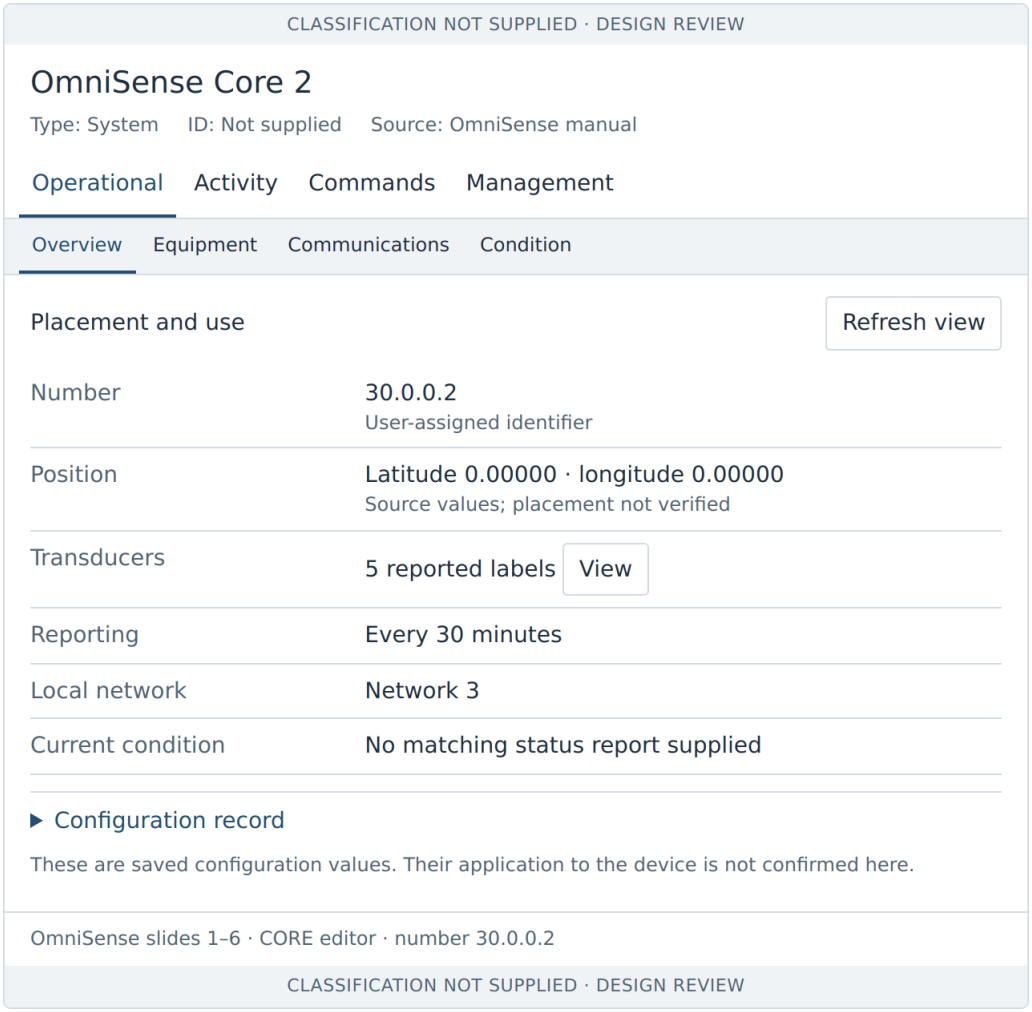
The first screen keeps the things Sentinel's General view put together: the source's state, position, string assignment, detector choices, radio setting and trigger relationship. These are immediately useful for access to this particular system. Sparse emplacement details remain one expansion away instead of occupying the default screen with empty fields. This follows your manual-first direction; it does not replace it with generic hardware health.

The labels also limit the claim. ANTICIPATED remains a Sentinel state, not a healthy or online indicator. Detector selections are configuration evidence, not identified attached resources. The trigger target remains a literal reference until identity can be resolved. The inactive date controls do not establish an emplacement or removal date. Exact source values are retained, including MM101 and the coordinate string.

A Deployment is still a possible organizing resource for a real emplacement or mission, but a string number alone does not establish that model. That is a binding decision to resolve, not a reason to omit the operational information.

OmniSense: the same front door, different evidence

Operational / Overview | CORE 2 configuration, number 30.0.0.2. [O1-6]



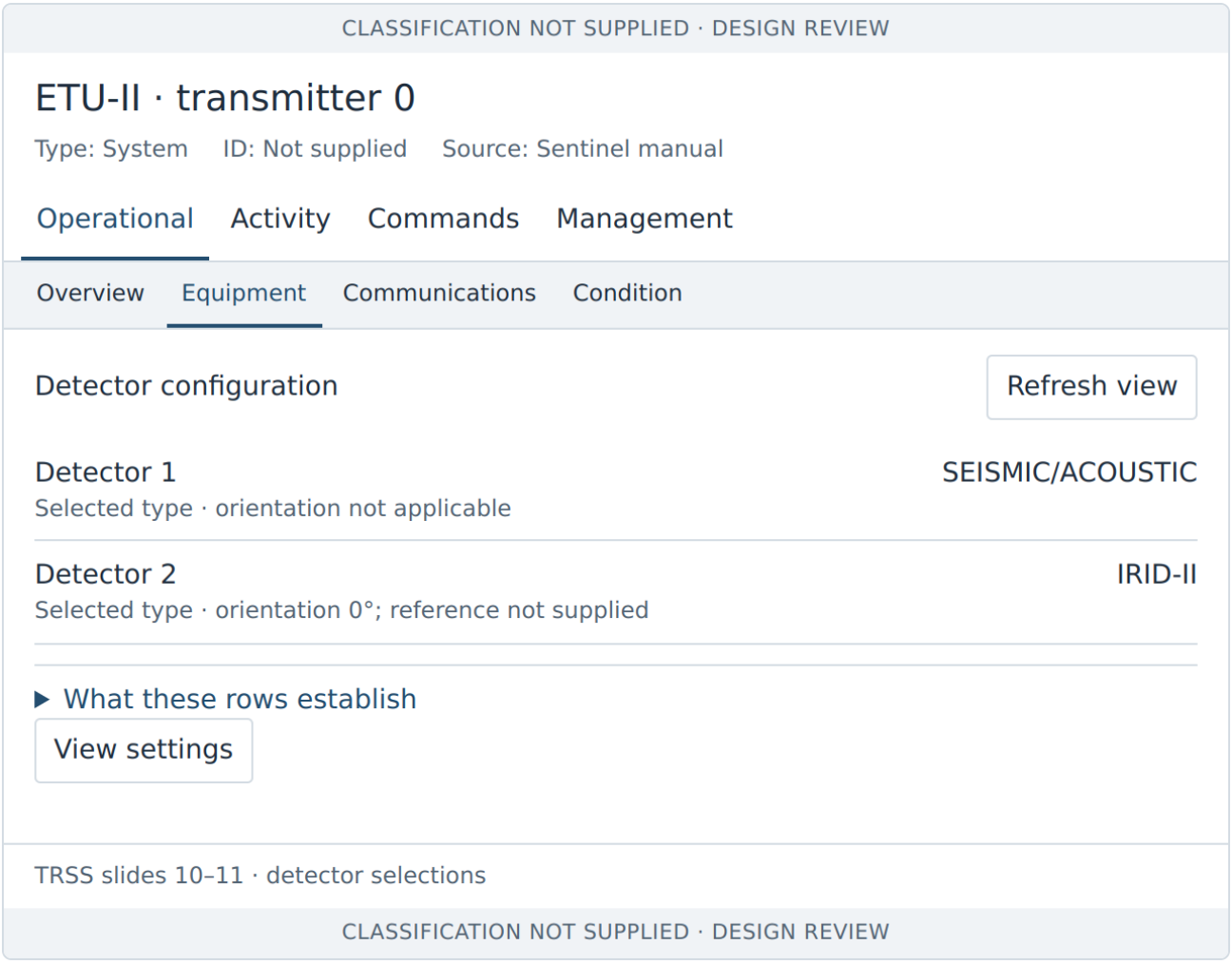
This uses the same compact frame and access pattern. The available evidence is different: a user-assigned number, location fields, reported transducer labels, reporting cadence and local network. It does not manufacture a sensor string or emplacement agency to match TRSS. Nor does the absence of those fields prove the actual deployed system lacks those relationships.

The zero coordinates are shown as source values whose placement is unverified. They are not silently converted to missing values or plotted as a confidently known location. The CORE 2 configuration pages do not supply a matching status report. The tempting voltage and green tests elsewhere in the manual belong to Core 5 ADS; importing them would make the wireframe look complete by giving it false evidence.

The manual defines Number and Description as operator-assigned identification text. Neither establishes a CSAPI ID or the generic technical description we discussed earlier. Firmware is visually transcribed here as 3.7.10.11. Any older transcription of this screenshot as 3.2.1011 should be corrected.

Equipment: configured detector positions

Operational / Equipment | TRSS General tab. [T10-11]



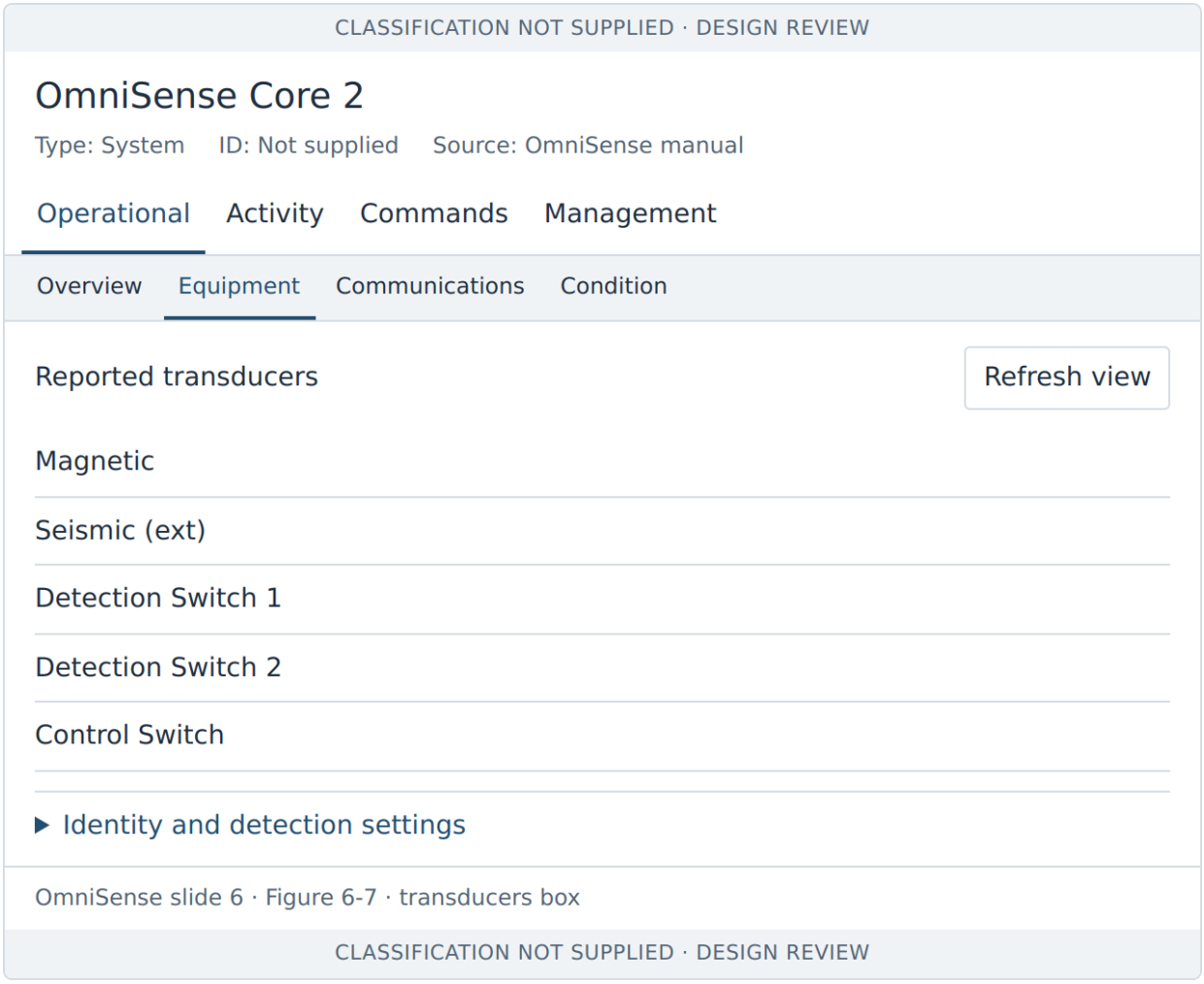
Equipment is a common destination because users need to understand what sensing equipment the system is set up to use. Here the manual gives two configured detector positions. Keeping those positions is useful and accurate. The visible selection is seismic/acoustic for Detector 1 and IRID-II for Detector 2, with an orientation value of zero for the latter.

This does not establish two independently identified sensor resources. The view therefore has no invented component links, serial numbers or attachment times. A stronger future binding could enrich the same view with those facts. It would also have to determine whether each relationship represents permanent composition or a time-qualified deployment association. An empty subsystem list would not alone settle that question. [D3]

The proposed carrier is typed configuration data. The detector definitions and allowable combinations remain part of a reviewed TRSS profile. The renderer can reuse rows and disclosures without knowing every vendor in its general layout code.

Equipment: a reported transducer list

Operational / Equipment | CORE 2 Transducers panel. [O6-7]



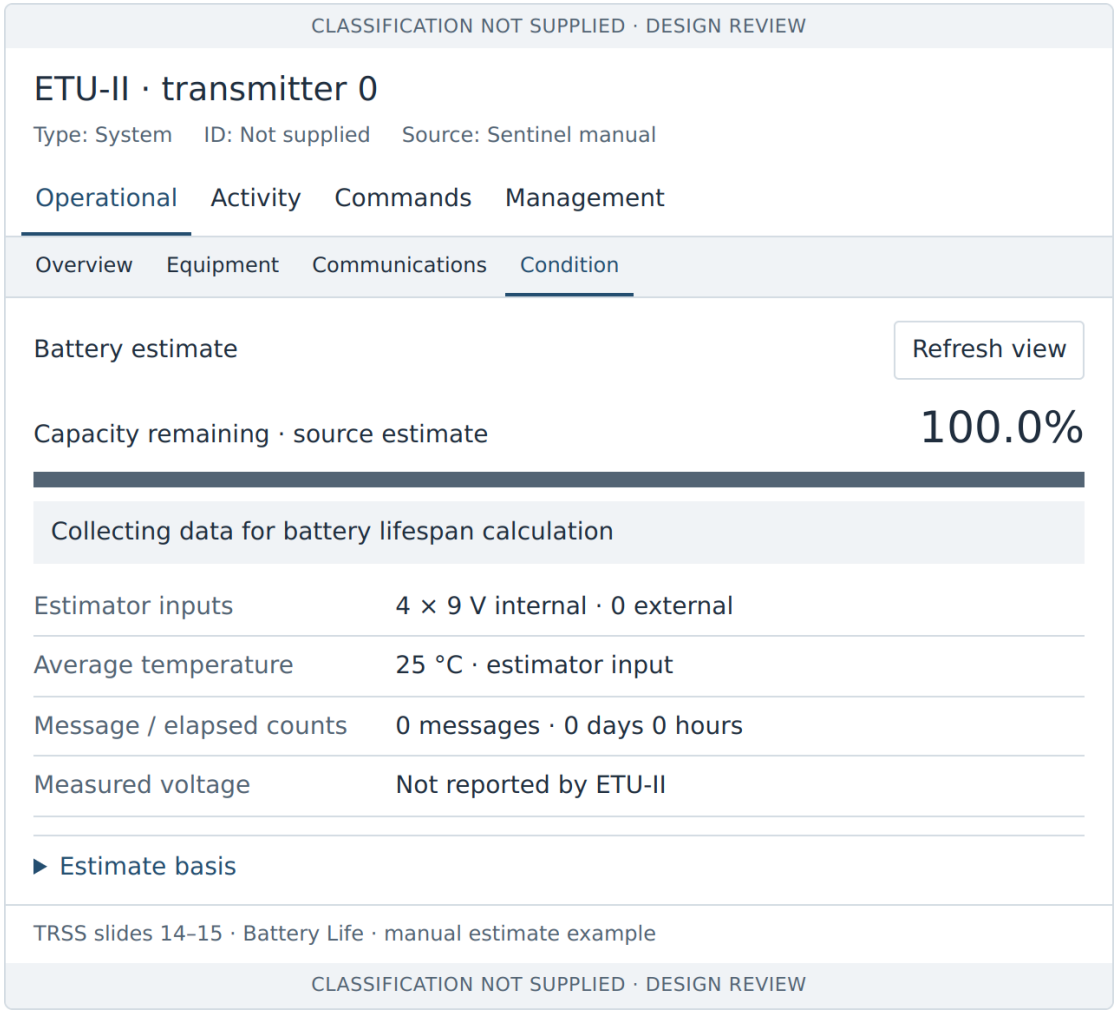
OmniSense describes automatic discovery of connected transducers during setup. Its screenshot supplies a list of five labels. This is different evidence from a manually selected detector type, so the shared Equipment view identifies the list as reported transducers instead of forcing it into Detector 1 and Detector 2 slots.

The list includes switch functions as well as sensing labels. Five labels should not be presented as five separately identified physical sensor resources. The screenshot supplies no component identifiers or connector assignments. A familiar row pattern can remain consistent while its evidence label preserves that difference.

The manual also separates transducers from detection algorithms: an enabled algorithm without its required transducer produces no detections. That distinction changes the design. Hardware information belongs here; sensitivity and algorithm settings belong in the read-first configuration view, with a shortcut between them.

Condition: an estimate must look like an estimate

Operational / Condition | ETU-II Battery Life. [T14-15]



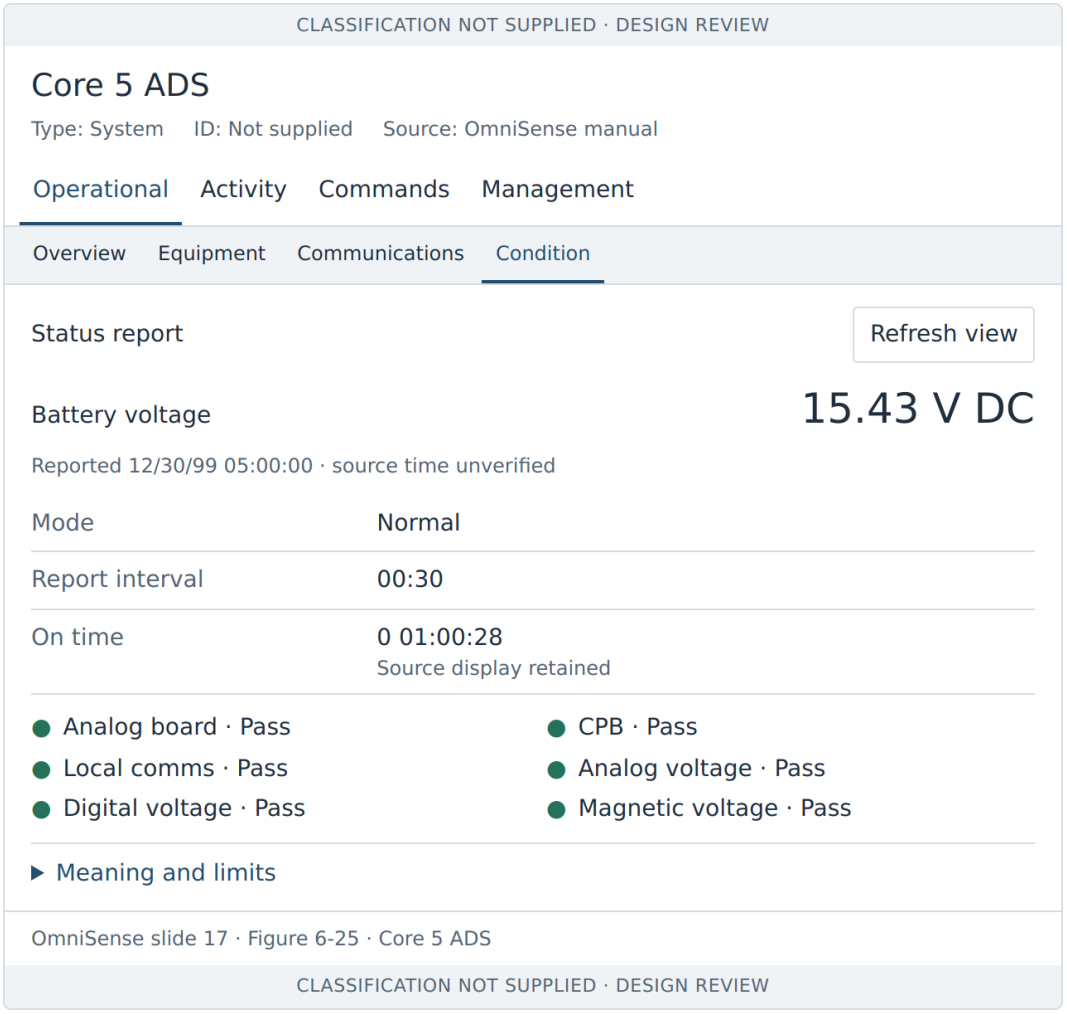
This preserves a visual indication, as you requested, but changes its meaning from a generic green battery to a neutral source estimate. Sentinel displays 100.0% while also saying it is collecting data. Both facts must survive. This view does not conclude that the battery is fully charged or that the system is ready.

The inputs explain why: four internal 9 V batteries, zero external batteries, an average-temperature input, and zero message and elapsed counts in the example. ETU-II explicitly does not return battery voltage. The manual says its estimate uses messages received by the SDR-II as a count of messages sent; missed receptions can therefore affect the calculation.

Condition is a useful common home. A universal state-of-charge field is not justified by these excerpts. The complete source/estimate schema, thresholds and finished visualization remain coordinated with the separate Power Source workstream; this pass does not supersede its decisions.

Condition: measured voltage and individual tests

Operational / Condition | Core 5 ADS, kept separate from CORE 2. [O17-18]



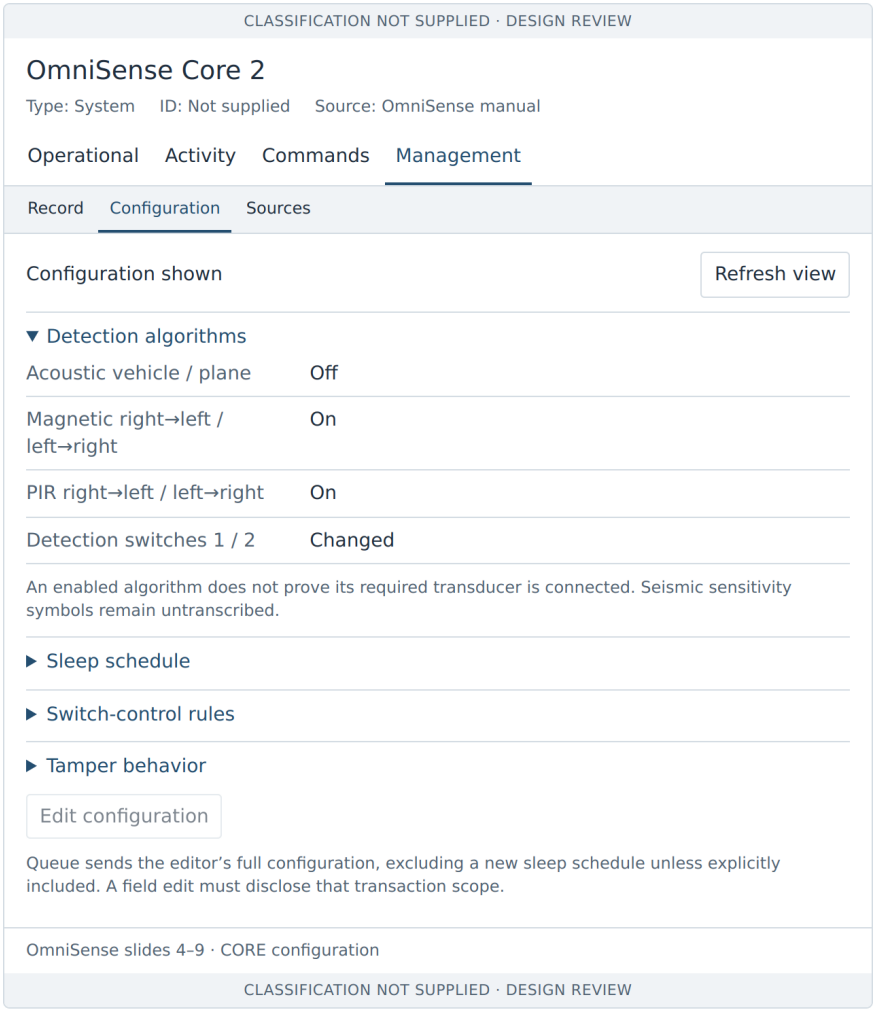
OmniSense supplies a measured voltage and a set of individual built-in test results. These deserve a different presentation from the TRSS estimator, inside the same destination. A large unit-qualified voltage is useful. A percent-full battery or remaining-runtime gauge would assert a conversion the evidence does not supply.

The six test passes remain separate facts. They do not prove overall mission readiness. The unusual 1999 timestamp is retained and identified as unverified rather than quietly made current. On Time is also preserved in the displayed source form because the prose calls it seconds while the screenshot uses a compound duration display.

The manual supports history tables and voltage charts. Those remain appropriate Activity views after the resource, schema, units, clock basis and series are bound. This selected screenshot is not enough to invent a trend, and samples for other CORE devices are not silently reassigned here.

Management: read first, edit with the right scope

Management / Configuration | CORE detection, schedule, switch and tamper groups. [O4-9]



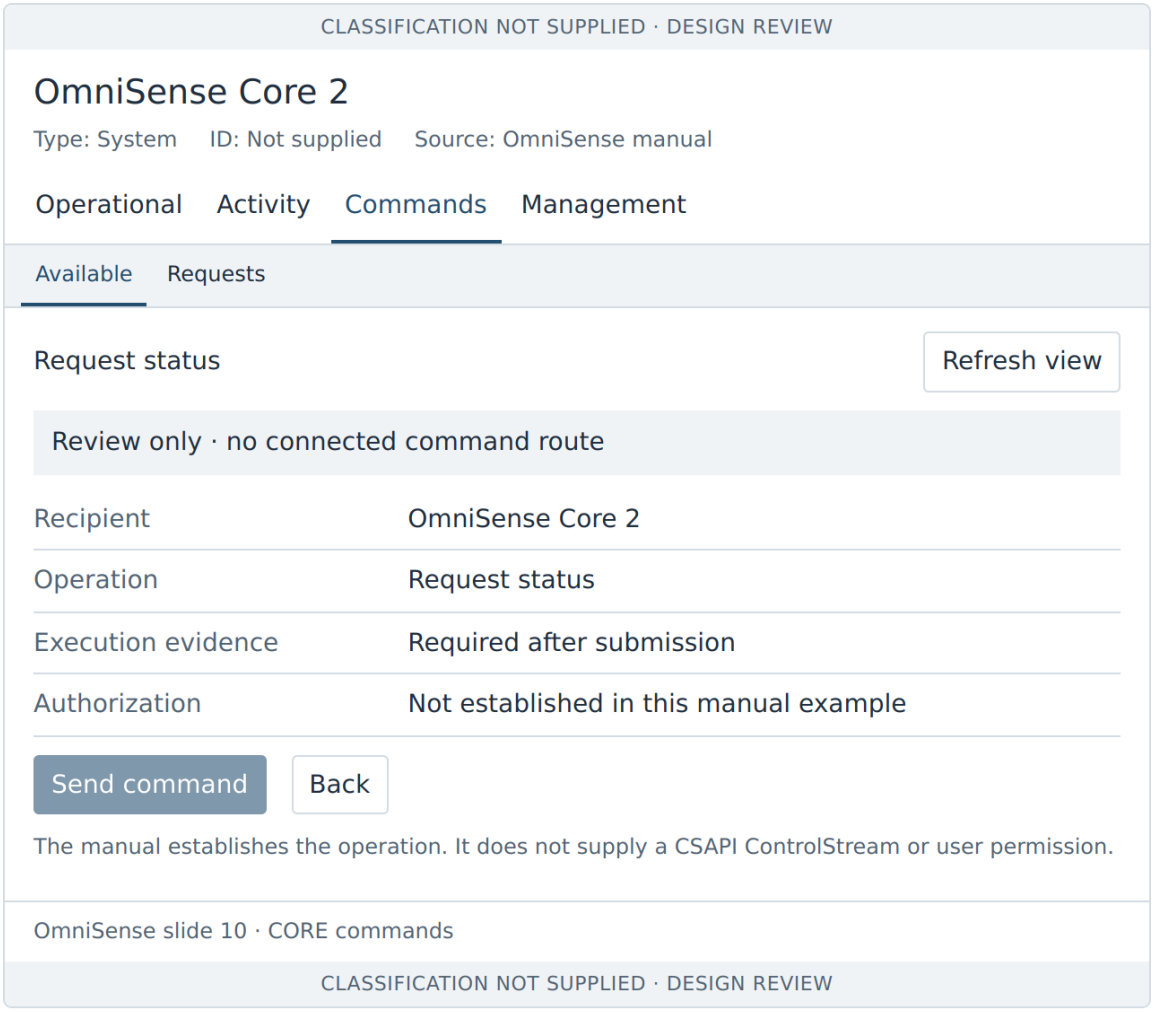
This is where useful structure from a standards-oriented editor can be retained without making it the entire front door. The initial experience is readable information. Disclosure keeps long groups contained, and the operational views can link to the same underlying settings. The repeated information has one binding and one authoritative edit workflow, not independent copies.

A crucial manual detail changes the edit design: Queue sends the editor's whole configuration, with the sleep schedule included only when explicitly selected. A client cannot honestly present every small change as an isolated write unless its adapter guarantees that behavior. A review must explain the actual transaction scope and distinguish a queued change from a device-applied change.

Tamper policies, weekly schedules and switch rules retain their own types and consequences. Their presence in SensorML/SWE does not make them interchangeable Boolean fields. The disabled Edit control in this review reflects the missing route and permissions, not a requirement that configuration can never be edited.

Commands: one workflow for different operations

Commands / Available | CORE request-status operation; no command is sent. [O10,20-21]



Commands remains a peer category because requesting an action is a different task from reading a value. The operation can still be reached from the relevant operational view. CORE documents requests for configuration, GPS location and status, plus Set Operational. ETU-II documents a Direct Configure operation. The available operations differ; recipient, scope, confirmation and outcome can follow a consistent interaction pattern.

The standard provides command and status structures, but a manual does not establish an actual ControlStream, parameter schema or user permission. The review is usable here; submission is unavailable. Sending, acknowledgment, application and completion must not be collapsed without a documented adapter mapping. [D4]

Activity can cross-reference the same request alongside observations and system events, with type filters. It should not clone the command workflow or relabel every vendor history row as an observation. The interactive draft preserves that common table structure but leaves unmatched histories empty rather than fabricating events.

What the standards carry, and what we still define

These are proposed representations, not asserted paths in an existing CSAPI response. Named leaf definitions must be published or mapped by a reviewed profile.

Manual concept	Available representation	Binding and presentation decision
Emplacing party and date	SensorML Event.time and Event.contacts[]; ResponsibleParty.organisationName and role. [D2a,D2b]	Use only for an identified actual event. Assign an emplacement definition and role; retain a literal party unless a resource identity resolves.
Deployment-specific notes	SensorML DeployedSystem.description, system and configuration. [D2c]	Only when the record actually describes participation in a deployment. Do not infer a Deployment from String Number.
Detector choices and tuning	SensorML parameters[] containing defined SWE components; configuration settings may reference parameter paths. [D2,D5]	TRSS detector choice and CORE algorithm settings remain different concepts. The profile chooses the common Equipment or Configuration destination.
Reported transducer labels	A defined repeated SWE record/list. [D5]	Bind source discovery semantics and time. Labels alone do not authorize component navigation.
Battery voltage / status report	DataStream schema.resultSchema and matching Observation.result, with source-qualified subject/time. [D4]	Separate voltage, test results, mode and duration. A quantity carrier does not invent thresholds or remaining runtime.
Request status / apply configuration	ControlStream schema.parametersSchema; Command and status/result representations. [D4]	Bind recipient, operation, permissions, transaction scope and outcomes. A view refresh is a different action.
System identity and firmware	Qualified SensorML identifiers and characteristics; RFC 8348 provides hardware identity/revision reference concepts. [D2,D6]	Do not replace API identity with a user label or collapse OS and application firmware into one ambiguous revision.

The practical client contract

Each promoted field needs: a versioned source/schema selector; its semantic definition; owning resource or component; value type, units and time basis; a supported display role; provenance; and any permitted read/write/command route. These are our proposed integration requirements. They are not extra fields claimed to be mandated by OGC.

Recognized shared definitions use common renderers. Reviewed vendor definitions use the same renderers where their meaning fits and specialist ones where it does not. Unknown definitions remain visible in readable source groups. This is the missing application contract; an OGC document’s validity alone does not supply it.

Coverage ledger: 01 onward

T = TRSS deck slide; O = SADU/OmniSense deck slide. E = shared concept/carrier; V = vendor binding required; U = interpretation unresolved. None means a tested adapter.

Subject / class	Manual evidence	Treatment, proposed home and unresolved work
01 Resource identity E	TRSS: ETU-II transmitter ID 0 [T10-11] CORE: CORE Number is user-assigned and may repeat [O1-2]	SensorML identifiers[] may preserve qualified aliases. CSAPI resource ID remains independently assigned. Home: Fixed header / Management · Record Open: Neither alias establishes an API ID; no payload supplied.
02 Name and description U	TRSS: General view has no generic product description [T10] CORE: Description is operator-assigned identification text [O2]	Do not populate a generic technical description from emplacement comments or an operator alias. Home: Title / Management · Record Open: Need a product/procedure description and an explicit precedence rule.
03 Horizontal position E	TRSS: 18STD4126589907 [T10-11] CORE: CORE 2 editor shows 0.00000 / 0.00000 [O3]	System feature geometry or a time-qualified position observation, according to source meaning. Preserve raw position until conversion is justified. Home: Operational · Overview; future Map Open: Datum, validity, time, configured-versus-measured authority; zero is not automatically null.
04 Height values U	TRSS: Elevation 0 m and Altitude 0 m are separate [T10-11] CORE: GPS altitude shown as 0.0 [O3]	Separate SWE quantities until vertical reference and meaning are established; no single generic height field. Home: Overview disclosure / Management Open: Vertical datum; why TRSS exposes both fields; placeholder versus real zero.
05 Sensor string and sketch V	TRSS: String MM101; Sketch Number blank [T10-11] CORE: No corresponding fields in CORE chapter [O1-23]	Defined placement/assignment properties; link only if an identified string or document resource exists. Home: Operational · Overview / emplacement disclosure Open: A string label alone does not create a Deployment or a document link.
06 Emplacing organization V	TRSS: Implant Agency; manual calls it Emplacement Agency [T10-11] CORE: No emplacement-party record supplied [O1-23]	Role-qualified ResponsibleParty associated with the actual emplacement Event; organisationName preserves a literal party. Home: Overview · Emplacement details Open: Event ownership/time and role URI required; a party name does not identify a System resource.
07 Actual emplacement date V	TRSS: Date placed; inactive calendar example [T10,12] CORE: Emplacement Mode is radio access behavior, not this date [O4]	A defined emplacement Event.time when an actual occurrence is known. Home: Overview · Emplacement details Open: No valid event can be manufactured from an unchecked date control or a mode label.
08 Planned removal V	TRSS: Intended removal date [T12] CORE: No counterpart supplied [O1-23]	Defined planning property, separate from an actual removal event and resource validity. Home: Overview · Emplacement details Open: Do not assume this is System.validTime or a Deployment end.

Coverage ledger: 09 onward

T = TRSS deck slide; O = SADU/OmniSense deck slide. E = shared concept/carrier; V = vendor binding required; U = interpretation unresolved. None means a tested adapter.

Subject / class	Manual evidence	Treatment, proposed home and unresolved work
09 Emplacement notes V	TRSS: Comments action; text not shown [T12] CORE: No matching notes supplied [O1-23]	Defined notes attached to their actual subject; DeployedSystem.description is a candidate only for a real deployment participation record. Home: Overview disclosure / Management record Open: Resolve owning record before editing; do not overwrite generic product description.
10 Vendor state V	TRSS: ANTICIPATED; manual warns against casual changes [T10-11] CORE: Normal in CORE 5 status [O17]	Distinct, defined Category values in their own source state models. Home: Operational · Overview / Condition Open: Neither is mapped to RFC oper-state, health or online without an explicit equivalence.
11 Operating / local mode V	TRSS: No ADS/CS equivalent established [T9-16] CORE: ADS or CS configuration; local Emplacement Mode [O4]	Separate SWE Category parameters; local access mode is not mission assignment or an emplacement event. Home: Overview when known / Management · Configuration Open: CORE 2 selected ADS/CS value is not visible; distinguish configuration from reported mode.
12 Detector positions V	TRSS: Detector 1 seismic/acoustic; Detector 2 IRID-II [T10-11] CORE: No equivalent numbered detector assignments [O6]	Typed configurable detector selections; portable equipment destination with vendor-specific row meaning. Home: Operational · Equipment Open: No identified attached instances or current connection confirmation.
13 Reported transducers V	TRSS: General choices are not an autodetected list [T10-11] CORE: Magnetic, Seismic (ext), two detection switches, control switch [O6]	A repeated, defined SWE record/list with reported labels and source basis. Home: Operational · Equipment Open: No IDs, serials, ports or one-to-one physical component count inferred from labels.
14 Replaceable attachments U	TRSS: Interchangeable detector types; instance identity absent [T10-11] CORE: Setup autodetection; identity absent [O6]	CSAPI composition or Deployment association after permanence, membership, participant identity and time are established. Home: Equipment / future related deployment Open: Manuals cannot settle existing server modeling or lifecycle support.
15 Detector orientation U	TRSS: 0 degrees for IRID-II; N/A for seismic/acoustic [T10-11] CORE: No corresponding orientation supplied [O6]	SWE Quantity for angle with a defined reference frame when known. Home: Operational · Equipment disclosure Open: Preserve numeric zero; no map bearing until direction reference is resolved.
16 Trigger relationship V	TRSS: Trigger for Imager-II ID 0 [T10-11] CORE: CS mode can cue a separate imager; correlated images [O4,11]	Explicit relationship and correlation evidence; navigate only after target identity resolves. Home: Overview / Activity related result Open: Do not give the CORE or ETU-II an intrinsic camera or invent a component relationship.

Coverage ledger: 17 onward

T = TRSS deck slide; O = SADU/OmniSense deck slide. E = shared concept/carrier; V = vendor binding required; U = interpretation unresolved. None means a tested adapter.

Subject / class	Manual evidence	Treatment, proposed home and unresolved work
17 Radio / network settings V	TRSS: TX Channel 350 [T10-11] CORE: Local Network 3 [O6]	Separate defined configuration parameters, not one interchangeable Channel value. Home: Operational · Communications Open: No frequency conversion or network membership identity without a binding.
18 Reporting cadence V	TRSS: Immediate directional messages or counters; hourly control disabled [T12-13] CORE: 30-minute configured report interval; separate reported interval [O5,17]	Enum reporting behavior and duration/cadence are separate parameters/results. Home: Communications / Management · Configuration Open: A disabled control does not establish support. Preserve configured and reported versions.
19 Neighbors and link counters V	TRSS: No neighbor-table counterpart supplied [T9-16] CORE: Up to 10 RF neighbors; RSSI and transmit/receive counts [O19]	Defined, time-qualified observation record/array with reported neighbor aliases and counters. Home: Communications; history under Activity Open: Resolve identity, counter periods and RSSI units; observed neighbors are not permanent children.
20 Detection tuning V	TRSS: Target masks and gain High/Medium/Low/Ext Only [T12-13] CORE: Separate algorithm sensitivity/on/off settings [O6-7]	Defined SWE parameter groups; retain per-algorithm constraints and applicability. Home: Management · Configuration; operational summary when useful Open: Do not normalize incompatible gain scales or equate enabled algorithms with attached sensors.
21 Tamper behavior V	TRSS: Enable detection; documented disabling effects [T13] CORE: Off, Report, Disable or Kill behaviors [O7]	Separate effect-bearing command/configuration definitions, not a shared Boolean. Home: Management · Configuration / command review Open: Route, privileges, irreversible effects and recovery method require explicit binding.
22 Sleep and delayed start V	TRSS: Power-on delay; start/stop sleep; detection and communications suspended [T13] CORE: Weekly day/start/duration schedule; HHPM needed to wake early [O8]	Separate typed schedule records with time basis, repetition and effects. Home: Management · Configuration; operational summary when active Open: Different schedule models; confirm configured versus effective state and timezone.
23 Switch rules V	TRSS: No counterpart established for ETU-II [T9-16] CORE: Classification + direction + open/close action [O9]	Defined rule DataRecord/DataArray or source schema; retain its structure. Home: Management · Configuration Open: Vendor semantics and validation needed; do not flatten into generic sensitivity.
24 Power-source configuration V	TRSS: 4 internal 9 V; external count 0; installation and temperature inputs [T14-15] CORE: No equivalent source inventory in selected CORE pages [O1-23]	Typed, defined source and estimator-input records; handoff to Power Source workstream. Home: Operational · Condition disclosure / Management Open: Selected external model is not an installed battery when count is zero.

Coverage ledger: 25 onward

T = TRSS deck slide; O = SADU/OmniSense deck slide. E = shared concept/carrier; V = vendor binding required; U = interpretation unresolved. None means a tested adapter.

Subject / class	Manual evidence	Treatment, proposed home and unresolved work
25 Estimated battery capacity V	TRSS: 100.0% while collecting data; 0 messages and elapsed time [T14-15] CORE: No capacity estimate in selected status [O17]	Defined estimate quantity with method, inputs and calculation status. Home: Operational · Condition Open: Not state of charge by assumption; reception-based message count can bias estimate.
26 Measured battery voltage E	TRSS: Explicitly not returned by ETU-II [T14] CORE: 15.43 V DC for Core 5 ADS; voltage history/graph [O17-20]	Unit-qualified observation Quantity associated with the correct system and sampling time. Home: Condition / Activity · Charts Open: No voltage-to-percent conversion, healthy threshold or CORE 2 attribution.
27 Built-in tests and uptime V	TRSS: No equivalent report in this ETU-II excerpt [T9-16] CORE: Six test passes and On Time in CORE 5 [O17-18]	Defined per-test categories/booleans and duration, preserving report time and test scope. Home: Operational · Condition; Activity history Open: No universal health score; resolve display duration versus manual seconds encoding.
28 Firmware E	TRSS: Operating-system and application versions distinct [T15] CORE: Manufacturer firmware field [O2]	Qualified firmware revision characteristics; RFC hardware model gives semantic precedent. Home: Management · Record Open: Do not erase multiple software roles; version display is not a firmware-update capability.
29 Detection activity V	TRSS: Immediate detections or counters [T12-13] CORE: Detection time, type, target ID; optional related image fields [O11]	Observation result schemas for distinct message semantics; preserve source time and correlation. Home: Activity · Table; optional feed later Open: No invented observed feature identity; counts and individual events need distinct schemas.
30 Detection charts E	TRSS: No chart dataset in the ETU-II excerpt [T9-16] CORE: Counts by type/hour/day [O12-16]	Client aggregation over an identified, time-bounded event dataset; aggregation is our behavior. Home: Activity · Charts Open: Time zones, duplicate handling, coverage and classification vocabulary must be explicit.
31 Configuration history V	TRSS: Configuration response messages mentioned [T16] CORE: Received time, queued update time, configuration snapshot [O21-22]	Preserve command association and configuration history; use SensorML Event only where its event meaning fits. Home: Activity · Table / Management · Configuration Open: Received, queued, applied and observed times are not synonyms.
32 Command requests and outcomes V	TRSS: Direct Configure, timeout and responses [T16] CORE: Configuration, GPS location, status, Set Operational [O10,20-21]	Verified ControlStream parametersSchema, Command and status/result resources; native evidence retained. Home: Commands · Available / Requests; Activity reference Open: No automatic Sent-Acknowledged → COMPLETED mapping; real route and permissions absent.
33 Configuration transaction scope V	TRSS: Local configuration transmission; manual warns against arbitrary changes [T9,16] CORE: Queue sends whole editor except sleep unless included [O2-9]	Review complete action scope; editing metadata and changing device configuration are distinct workflows. Home: Read-first Management / Commands review Open: Single-field UI edits must not conceal a whole-configuration write; concurrency policy needed.

Coverage ledger: 34 onward

T = TRSS deck slide; O = SADU/OmniSense deck slide. E = shared concept/carrier; V = vendor binding required; U = interpretation unresolved. None means a tested adapter.

Subject / class	Manual evidence	Treatment, proposed home and unresolved work
34 Refresh and live behavior E	TRSS: Device messages separate from the local view [T16] CORE: History refreshed by interval selection; request status is an operation [O10-11,17-18]	Separate reread, polling/stream updates and command-driven acquisition. Home: View toolbar / Activity / Commands Open: No implied live route; preserve data age after refresh.
35 Export and sharing U	TRSS: Not established in the selected ETU-II excerpt CORE: Clipboard, print, file and IBS export [O11-21]	Client export workflow plus target format and security policy, not device configuration. Home: Activity export; later PBI slice Open: IBS format, marking transformations and release checks need separate requirements.
36 Display and alarm preferences V	TRSS: Not established in this excerpt CORE: Visible/audio alerts, ranges, neighbors, image popup [O23]	Client preferences/subscriptions unless an explicit source contract says otherwise. Home: View preferences; outside device parameter editing Open: Do not send map-display preferences to the physical sensor or disable its detection by inference.
37 Remove from map V	TRSS: Not established in this excerpt CORE: Removes icon; record retained; reporting can restore it [O22]	An application visibility action with documented sharing effects. Home: Management/view actions, deferred Open: Not CSAPI DELETE, decommissioning, removal from a deployment or a device command.
38 Security markings U	TRSS: Manual excerpts, not per-resource security metadata CORE: IBS export shows classification choices [O17]	Security metadata and policy bind banners/portion markings; SensorML securityConstraints alone does not complete the UI policy. Home: Fixed top/bottom banners and marked portions Open: No field markings supplied; prototype explicitly leaves classification unknown.
39 Symbol, map, media and contacts U	TRSS: No bound symbol/profile or media/contact records in this case CORE: Map and correlated imagery described; no complete bindings [O3,11,23]	Existing module work remains in scope; activate views with actual definitions, identities and resources. Home: Map candidate under Operational; Symbol placement still open Open: This comparison does not settle symbology placement or retire these capabilities.
40 Unknown vendor information E	TRSS: Future fields and unfamiliar codes CORE: Future fields and unfamiliar codes	Readable SensorML/SWE structure; promote only after reviewed semantic binding. Home: Management source groups; no silent omission Open: A valid carrier alone does not establish meaning, editability or a custom renderer.

What is established enough to build next

Recommendation: review the two Operational / Overview examples, then implement one bounded, data-backed slice.

Candidate requirements from this pass

One information architecture

Keep stable destinations and two tab levels. Vary the fields by explicit semantic applicability, not vendor-name conditionals scattered through the UI. The proposed labels still need your review.

Evidence stays attached

Every displayed value retains its owner, basis and time. Configuration, reports, estimates and client preferences must not be mixed. Do not move CORE 5 status onto CORE 2.

Placement is useful without forced equivalence

Show established assignment/emplacement information, but do not create a Deployment, party-resource link, global ID or height meaning from an ambiguous label.

Readable before editable

Reuse one underlying record and edit path across shortcuts. Explain whole-configuration writes and preserve queued versus applied state. Show commands only under the real route/permission contract.

Gaps remain legible

Unsupported fields are not zero, failed or healthy by default. Retain unknown information in readable source groups; do not invent a map, chart, component or security marking.

Verification and its boundary

The interactive draft passed 117 local assertions across its content distinctions, interactions and layouts at 320, 640 and 736 CSS pixels. The important checks include device separation, disabled unbound writes, numeric zero retention, and refresh preserving source time. The ledger accounts for 40 subjects. This is UI and evidence-accounting verification; it is not new OGC conformance testing or a live sensor test.

The next engineering input is one real System description, one relevant DataStream/schema and observation, and any applicable ControlStream/schema for each selected device. Have the integration owner supply those exact records; more screenshots are not the missing ingredient for this test. Start with Equipment: the same view must handle TRSS choices, CORE labels, an identified component, and an unknown field without changing their meanings.

Sources and continuity

Primary documents and supplied excerpts used directly in this iteration. URLs are live references; cited versions identify the researched baseline.

T Supplied TRSS Details View Research.pptx, slides 9-16. Sentinel 1.0.7.2 Software Users Manual, N65236-TRSS-SUM-0005-2.00; General, Advanced, Battery Life, Firmware, Direct Configure. Slides 17-18 were checked as the EUGS boundary and excluded from ETU-II facts.

O Supplied SADU Details View Research.pptx, slides 1-23. OmniSense System User’s Manual Central Computer/Server and User Interface, © McQ 2008; printed pages 6-1 through 6-23. CORE 2 editor, CORE 5 status and other history examples are distinct.

D1 OPC UA Part 100: Devices, v1.05.0, §§4.4.1-4.4.2. FunctionalGroup model and well-known purposes. Used as organization precedent, not a required OGC transport or completed crosswalk.

https://reference.opcfoundation.org/specs/OPC-10000-100/4.4.2

D2 OGC SensorML 3.0, OGC 23-000, §§8.2, 8.8-8.9 and JSON implementation. Descriptive carriers, configurable processes and deployment participation.

https://docs.ogc.org/is/23-000/23-000.html

D2a Official Event JSON schema: required label/time, optional definition, contacts, properties and configuration settings.

https://schemas.opengis.net/sensorML/3.0/json/Event.json

D2b Official ResponsibleParty JSON schema: role and party identity.

https://schemas.opengis.net/sensorML/3.0/json/ResponsibleParty.json

D2c Official DeployedSystem JSON schema: description, system, configuration. The UML name is not copied as an invented direct JSON field.

https://schemas.opengis.net/sensorML/3.0/json/DeployedSystem.json

References and remaining scope

The additional standards are used selectively, where they change the design or prevent a misleading equivalence.

D3 OGC API - Connected Systems Part 1, OGC 23-001, §10.2 and subsystem requirements. Permanent composition and changeable aggregation are distinct modeling considerations.

https://docs.ogc.org/is/23-001/23-001.html

D4 OGC API - Connected Systems Part 2, OGC 23-002, schemas, observations, controlstreams and commands; especially §§16.1.4-16.1.9.

https://docs.ogc.org/is/23-002/23-002.html

D5 OGC SWE Common Data Model 3.0, OGC 24-014. Typed values and structured data provide carriers; vendor definitions still need a semantic contract.

https://docs.ogc.org/is/24-014/24-014.html

D6 IETF RFC 8348, A YANG Data Model for Hardware Management, 2018, §§3 and 7. Identity, revisions and state distinctions. No automatic mapping from ANTICIPATED or Normal to its state model is claimed.

https://www.rfc-editor.org/rfc/rfc8348.html

What this changes in the existing work

This addendum extends the working-views packet; pages 1-16 remain intact. It brings the two selected manual families into one explicit organization, separates CORE instances, corrects the CORE 2 firmware transcription for the inspected screenshot, and makes the remaining semantic and integration gaps reviewable. It does not amend the master research paper, semantics guide, requirements baseline or separate Power Source decisions by implication.

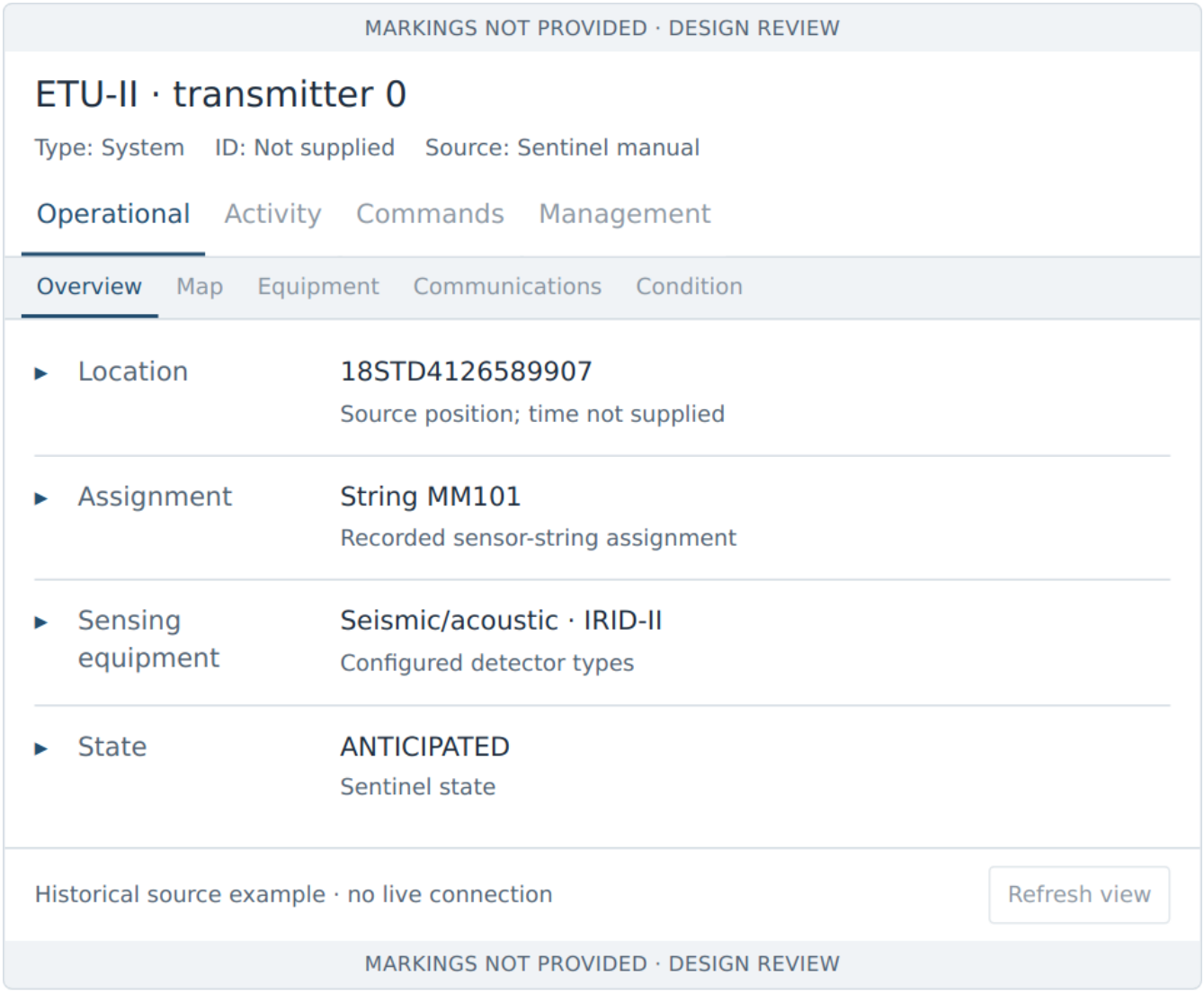
What is deliberately still open

Map placement, Symbol and its standard-specific subtabs, portion-marking rules, contacts/images, full deployment lifecycle, detailed Activity charts/feed/live behavior, additional device models, and final dimensions remain continuing PBI work. This research prototype targets a 640 CSS-pixel-wide card, 14-pixel body text and 12-pixel supporting text; content determines height. Those are review dimensions, not locked product requirements.

The OMA Temperature example on earlier pages remains a separate demonstration of a domain model with specific reusable semantics. It is not cited as a solution to emplacement, radio settings or CORE power. Likewise this packet does not add or reinterpret DISR status claims. Adoption and profile decisions must remain tied to the actual DISR entries and versions already researched.

Operational / Overview: TRSS

A narrower review of four priorities within the existing proposal. Other tabs remain provisional; this is not a replacement for the full module.



Why these four rows

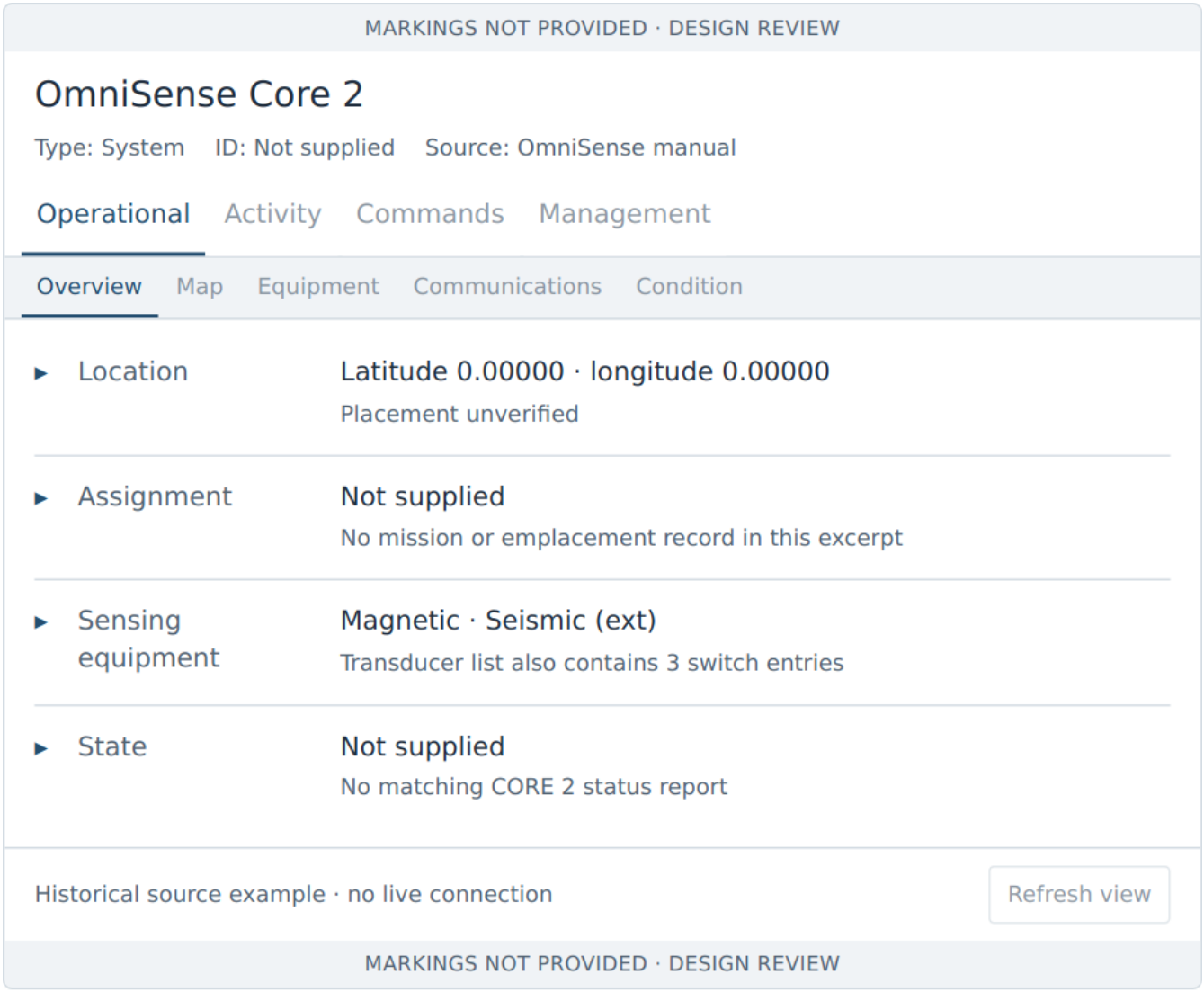
The Sentinel General view puts position, sensor string, detector selections and state close to the operator. The older MASBUS wireframes also foreground placement, assignment and components. We propose those as this UGS overview's first questions, rather than elevating the vendor configuration form. These are review priorities, not four mandatory fields for every kind of System. [F1, F4]

The qualification below each value matters. The manual describes detector types to be attached; it does not establish two identified installed resources. ANTICIPATED remains a Sentinel state, with no invented green health badge. A chevron expands detail in place; it does not navigate away. Transmitter 0 is a vendor identifier, not a confirmed API ID. [F1]

The 640 CSS-pixel card, fixed Type / ID / Source, two tab rows, refresh location and marking-banner positions are retained. Banner text is a review placeholder; no classification has been inferred.

The same overview: OmniSense Core 2

Same destinations; source-specific meaning remains visible. This excerpt concerns CORE 2, not Core 5 ADS.



Consistency does not require pretending the values match

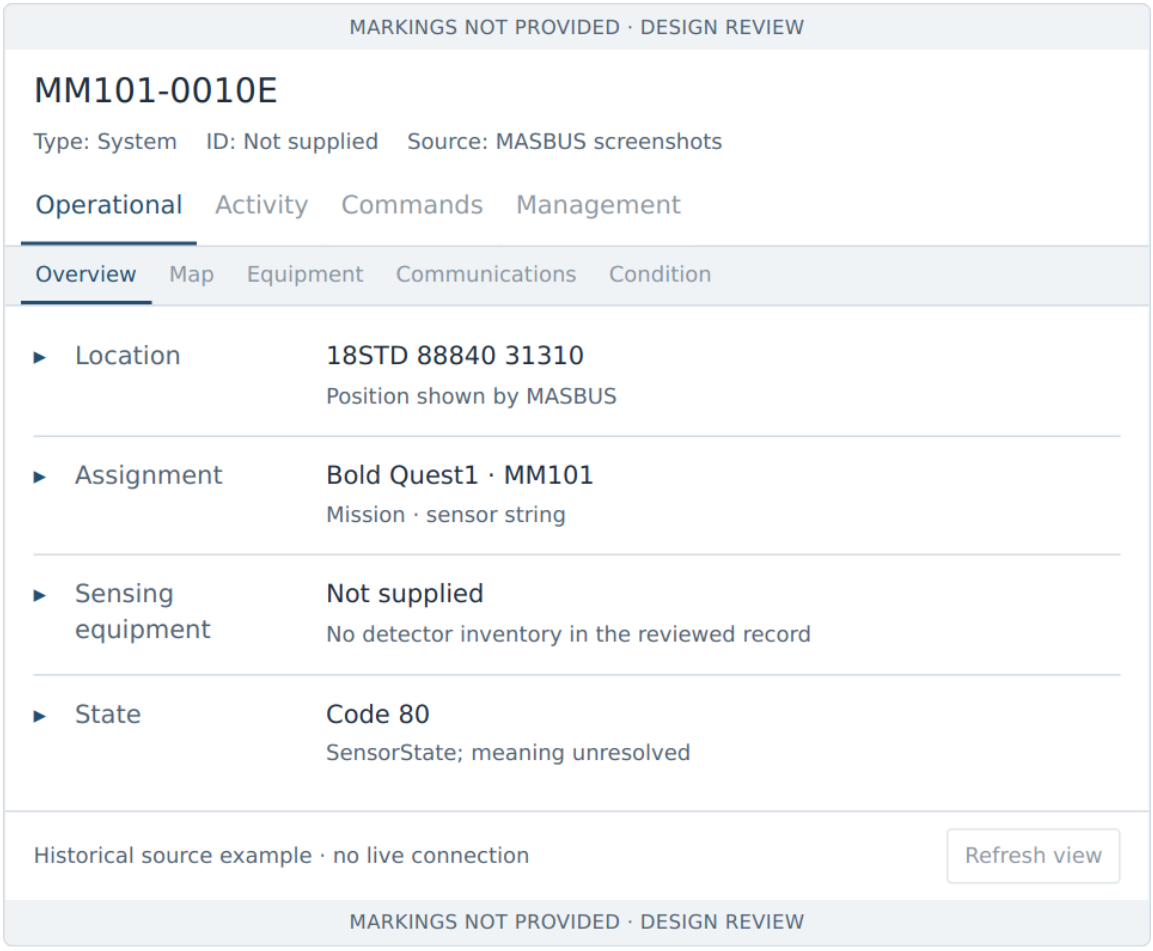
The CORE 2 Location editor contains zero coordinates. We keep those values visible but do not claim a verified placement. Its transducer list includes Magnetic, Seismic (ext), two Detection Switch entries and Control Switch; the expansion retains all five. The manual describes detection and reporting during setup. That is different evidence from TRSS's configured detector selections. Neither supplies individual resource identities. [F2]

Assignment and state remain explicitly unavailable for this selected example. Local Network 3 belongs to communications, and Emplacement Mode describes radio-access behavior; neither is a substitute for a mission or emplacement record. The later Core 5 status example cannot supply CORE 2's state. This prevents a misleadingly complete card. [F2]

For this comparison, unavailable rows stay visible so the reviewer can inspect the gaps. Final rules for hiding optional, inapplicable sections remain a separate decision; unavailable is not the same as inapplicable.

MASBUS supplies a concrete source anchor

A third, distinct record: MM101-0010E. Sharing the string MM101 does not make this the manual's transmitter 0.



What the implementation actually exposes

Visible carrier in the full description [F3]	Value / treatment
Identifier definition: urn:ogc:def:identifier:DSOSRI:1.0:sensorString	MM101; display as a sensor-string label.
Identifier definitions ending missionName and planName, with the same DSOSRI:1.0 prefix	Both show Bold Quest1. Show the mission once; retain Plan in the expansion.
Characteristics: SensorState	80; preserve the code, with meaning unresolved.
Point(EPSG:4979)	34.6019248962402, -77.3027191162109, 0.0; preserve reference-system and axis-order evidence.

These are transcriptions from rendered screenshots, not inspected API payloads. The legacy definition strings are evidence of the old implementation; this review does not certify them as registered OGC definitions. The default view also shows AOI CombatTown, retained in the expansion. No detector inventory is supplied by this record. [F3]

From source facts to standards carriers

Carrier = where a standard can represent the information. Binding = the exact definition, subject and field the client is authorized to interpret.

Overview fact	Standards carrier / design use	What must still be established
Location	CSAPI: System location; SensorML JSON /position or GeoJSON /geometry. The API distinguishes current/latest location from a requested historical snapshot. [S1]	Authoritative position, CRS, conversion, axis order and position time. CORE editor values must not silently replace a confirmed position.
Assignment	SensorML: /identifiers can carry qualified identifiers; the official schema explicitly includes mission ID as an example. /identifiers is not a deployment-membership relation. [S2]	Define sensor-string and mission terms and their authority. MASBUS gives legacy evidence; TRSS does not expose its serialized carrier. Link a Deployment only when the actual association exists.
Sensing equipment	Proposal: TRSS detector choices as defined SWE Category components in SensorML /parameters. A CORE reported list would be read through a declared result schema. [S3, S4]	Actual property definitions, slot semantics and producer schemas. Separately identify attached resources before presenting linked equipment inventory.
State	Source-preserving display: retain the source code or category. RFC 8348 separates administrative, operational, usage, alarm and standby states; it does not decode these examples. [S6]	Full code vocabulary, owning subject, state time and lifecycle semantics. SensorState 80 and ANTICIPATED cannot be normalized by label matching.

Where the newer device research changes the design

OPC UA Devices FunctionalGroups allow information to be organized for application purposes, with recognized groupings such as Operational and Status. We use that as precedent for a purpose-based overview drawing from the full record. It does not prescribe these four rows, mandate our tab names or make our client OPC UA compliant. [S5]

For replaceable sensors, CSAPI distinguishes permanent composition from changeable aggregation and describes using Deployments for the latter. This is a concrete modeling direction, not a missing concept in the standard. The unresolved part here is the instance evidence: which identified sensors, associated with which core, for which period. [S1]

Requirements proposed by this review

Bounded implementation rules for this overview. They remain proposed requirements; the user has not locked the row selection or all category labels.

1. Preserve the compact shell

Keep resource Type, API ID and Source fixed; retain top/bottom marking locations and adjacent category/subcategory tabs. Use a native disclosure control for each expandable fact. Never substitute a vendor number for an API ID without a documented mapping.

2. Bind by meaning, not by a familiar label

Each recognized field requires a reviewed definition or source mapping, its owning subject, type/units, version or vocabulary scope, time basis and intended display role. A label containing "status" or "emplacement" is insufficient.

3. Show the evidence basis

Distinguish configured, reported, installed, estimated and user-entered values. Do not merge desired configuration with confirmed application, or use a record-validity date as the observation or emplacement time.

4. Keep ambiguity useful and honest

Retain unknown codes and their source labels. Keep unavailable, inapplicable, withheld and unsupported states distinct when the server provides that distinction. A blank field never implies healthy, zero or unassigned.

5. Link only established resources

A sensor-string literal may be useful before Deployment modeling is complete. Resource navigation requires a resolved identity and a verified relationship; detector labels alone must not become fabricated subsystems.

6. Share bindings across views

An overview shortcut and the full Management record must refer to the same underlying fact. Readable detail comes first; an edit or command is offered only under the correct supported route and permissions. This review establishes no write endpoints.

7. Keep time and refresh separate

Refresh view retrieves available data; it does not by itself command the device or make old data fresh. The source time, when known, remains visible. The historical prototype therefore disables refresh.

8. Preserve the broader scope

Communications, condition/power, activity, commands, symbols, raw representations and other prior work are not removed by this four-fact exercise. Power source remains its separate standards/profile workstream.

Verification: 16 local interaction/content/layout checks passed, including separation of CORE instances, retention of zero, unknown state handling and widths from 320 to 736 CSS pixels. Light and dark renderings were inspected. This validates the mockup, not an adapter, sensor integration or OGC conformance.

Evidence, limits and the next review

The new section is pages 35-40. Pages 1-34 and their attachments remain intact; this update changes the working-views packet only.

Supplied evidence

F1. TRSS Details View Research.pptx, slides 10-12. Sentinel 1.0.7.2 Software Users Manual, N65236-TRSS-SUM-0005-2.00. General view and explanations of detector selections, state, sensor string, position and emplacement controls.

F2. SADU Details View Research.pptx, slides 1-6 (manual pages 6-1 through 6-6), CORE 2. Slide 17 is a different Core 5 ADS status example and is excluded from CORE 2 values.

F3. MASBUS Details View.pptx, slides 1-3. Default details and full-description screenshots for MM101-0010E; identifier definitions, characteristics and position were visually inspected.

F4. UGS Client Development Wireframes.pptx, slides 6-7. TRSS and OmniSense designs support the proposed operational priorities. They are design examples, not observed device inventories.

Primary standards used in this section

S1. OGC API - Connected Systems Part 1 v1.0, OGC 23-001: §§9.2.3, 10.2, 10.5 and 19; location encodings and system associations.
https://docs.ogc.org/is/23-001/23-001.html

S2. SensorML 3.0 official DescribedObject JSON schema: identifiers and characteristics.
https://schemas.opengis.net/sensorML/3.0/json/DescribedObject.json

S3. SensorML 3.0, OGC 23-000: identifiers, parameters, settings and deployments; see also official AbstractProcess JSON schema.
https://docs.ogc.org/is/23-000/23-000.html

S4. OGC API - Connected Systems Part 2, OGC 23-002: DataStream schemas and observations.
https://docs.ogc.org/is/23-002/23-002.html

S5. OPC UA Part 100: Devices, §4.4 FunctionalGroupType. A presentation-organizing precedent, not a mandated OGC UI.
https://reference.opcfoundation.org/specs/OPC-10000-100/4.4

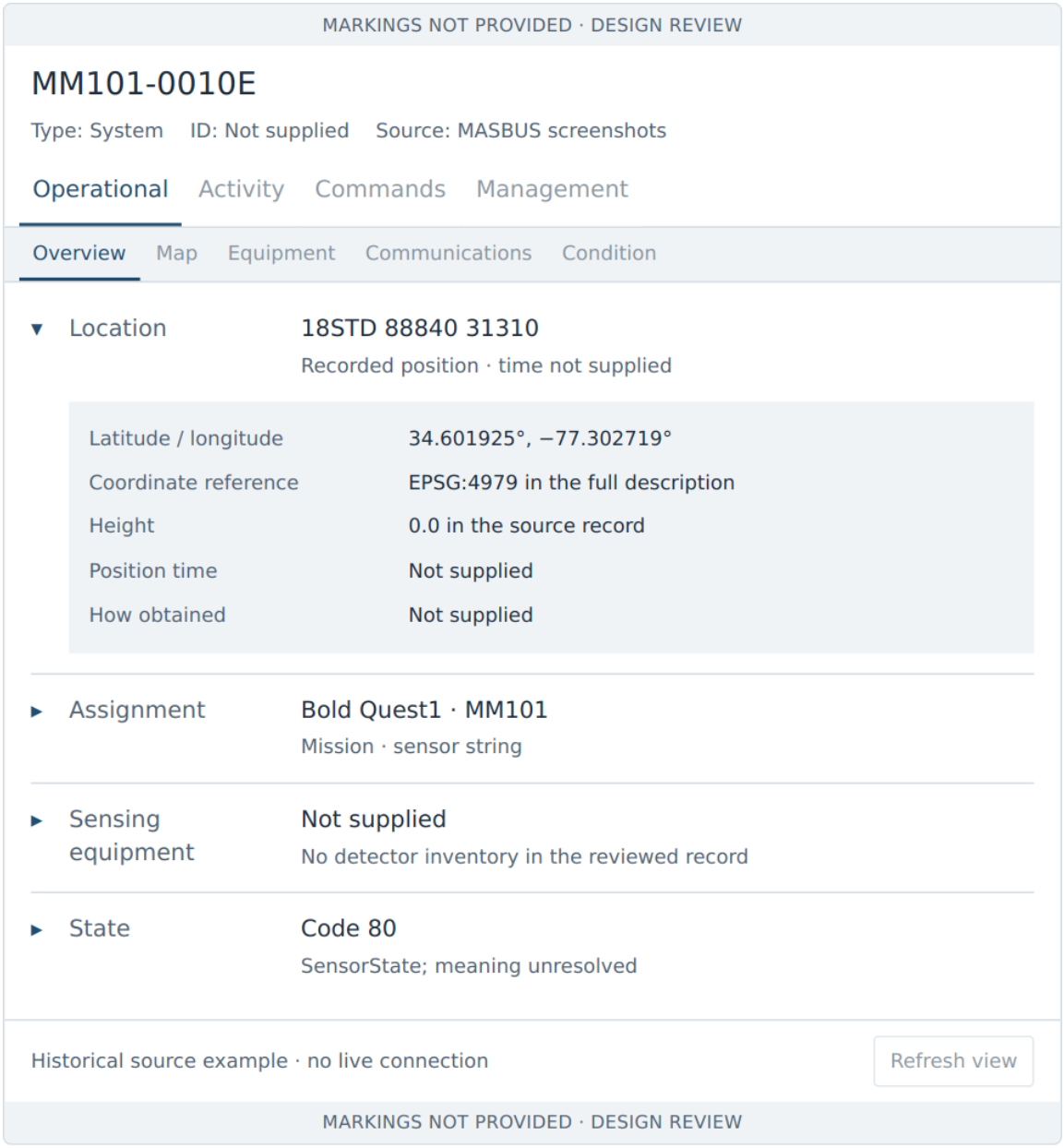
S6. IETF RFC 8348: hardware state distinctions. No asserted mapping for either source state in this review.
https://www.rfc-editor.org/rfc/rfc8348.html

Next: review Location, then Assignment, then Sensing equipment, then State inside this one overview. Agree on meaning and priority before changing more tabs. Payloads will validate the proposed bindings when available; they are not a prerequisite to reviewing this design. No universal sensor-state mapping, full deployment implementation or new DISR status determination is claimed.

Embedded supporting files: operational_overview_review_facts.json (transcriptions and proposed bindings, not API payloads) and operational_overview_verification.json (local UI checks). The master research paper and semantics guide have not been amended by this packet update.

Location: one row, two levels of detail

Proposed meaning: the location of the selected system at the time being viewed. The review opens Location; the normal overview can start collapsed.



The coordinate remains the main value. The small line tells the user whether its time is known. The chevron exposes supporting detail without leaving the card or opening a second page. Numeric coordinates should wrap when necessary, never be shortened with an ellipsis. A user's coordinate-format preference changes presentation, not the underlying location.

This example preserves MASBUS's MGRS display. Decimal coordinates are rounded to six places for readability; this is not a claim of measurement accuracy. Height remains the recorded 0.0, with the source CRS visible. The screenshot does not establish a position time or acquisition method. [F3, pp. 37, 40]

What comes from the standards

Keep the standardized location separate from optional information about its measurement, provenance and display.

Displayed item	Meaning and binding	Presentation rule proposed here
Location	CSAPI System location: latest known, unless a snapshot is requested. GeoJSON: /geometry; SensorML: /position. [L1]	Bind this row to the selected System. Do not substitute its target, observed feature, deployment area or a parent's coordinate merely because those are available.
Coordinate format	GeoJSON point coordinates use longitude, latitude, optional height. MGRS is a display conversion or preserved source notation. [L2]	Keep the source CRS and axis order. A correct coordinate transformation is required before treating another representation as equivalent.
Height	A vertical value needs units and a reference. GeoJSON's optional third coordinate is height in metres relative to the WGS 84 ellipsoid. [L2]	Do not merge altitude, elevation, height above ground or depth solely by label. Retain unresolved source fields separately.
Position time	Optional supporting information with an explicitly bound source. System validTime describes the system-description validity interval. [L1]	Do not automatically reuse validTime, a fetch timestamp or a generic Last Updated value as the time the position was measured.
How obtained / accuracy	Optional enrichment; no universal acquisition-method or accuracy leaf is established by this review.	Display a supplied method, uncertainty or accuracy only after its definition and units are understood. Do not infer GPS from editability or accuracy from decimal places.

An important boundary between SensorML and the API profile

The broader SensorML position schema supports text, a point, a pose, a process or a datastream link. Those possibilities are not all interchangeable with the CSAPI System location mapping. [L3] The first implementation should explicitly support its agreed encodings and expose unsupported positional content readably, without inventing a geographic point.

A useful location does not require every supporting field

A valid point can be displayed and mapped even when its measurement time or acquisition method is absent. Use "Time not supplied" where appropriate. Reserve "Last known" for a server or integration contract that establishes that role, and "Live" for an actually supported update mode. Staleness thresholds require a separate, resource-appropriate policy.

Location requirements ready for review

This pass refines the Location expansion only. Category names, other overview rows and the wider PBI remain under review.

Source example	What this review can use	What it must preserve
TRSS General view	Recorded sensor position; distinct elevation and altitude fields. [F1]	Original MGRS text; unresolved datum/height meanings; unknown position time. Editability does not prove manual entry.
OmniSense Location panel	GPS-based and subsequently stored location information; fields can be edited and queued. [F2]	The supplied zeros remain visible. Zero is not a missing-value test. Pending edits and subsequently reported values have different roles.
MASBUS details	MGRS display and full-description EPSG:4979 tuple for the same named record. [F3]	Original precision and CRS; the screenshot's two representations were not independently reconciled by a conversion test.

LOC-01. Identity and time

Use the selected system and requested time. A parent location requires an explicit derivation rule, with its basis retained. Description validity and position time remain separate.

LOC-02. Display and expansion

Keep the coordinate in the collapsed row, with a short time/quality qualification. Expand supporting details in place. Optional information does not have to produce permanent blank rows; this review exposes the gaps for discussion.

LOC-03. Map and copy actions

Propose View on map as a shortcut to Operational > Map for the same resource/time, using the same normalized point. Enable copy for the displayed coordinate. Do not require an acquisition timestamp merely to map a usable point. These actions are specified here, not implemented in this focused prototype.

LOC-04. Editing and refresh

Offer editing only through the supported route: record correction, report submission or device command, as appropriate. Keep pending values distinct from confirmed values. Refreshing the view does not itself request a new GPS fix.

LOC-05. Honest fallback

Preserve unknown reference systems and unsupported relative positions readably. Do not silently geocode a description, place an unknown system at 0,0, or plot a relative offset as an absolute coordinate.

Evidence: F1-F3 are the supplied sources identified on page 40; OmniSense page 6-3 was re-inspected for its GPS/database/Queue workflow. L1: OGC 23-001, §9.2.3 and Tables 4, 40, 49. L2: RFC 7946, §§3.1.1 and 4. L3: official SensorML 3.0 AbstractPhysicalProcess.json, $defs/Position.

https://docs.ogc.org/is/23-001/23-001.html

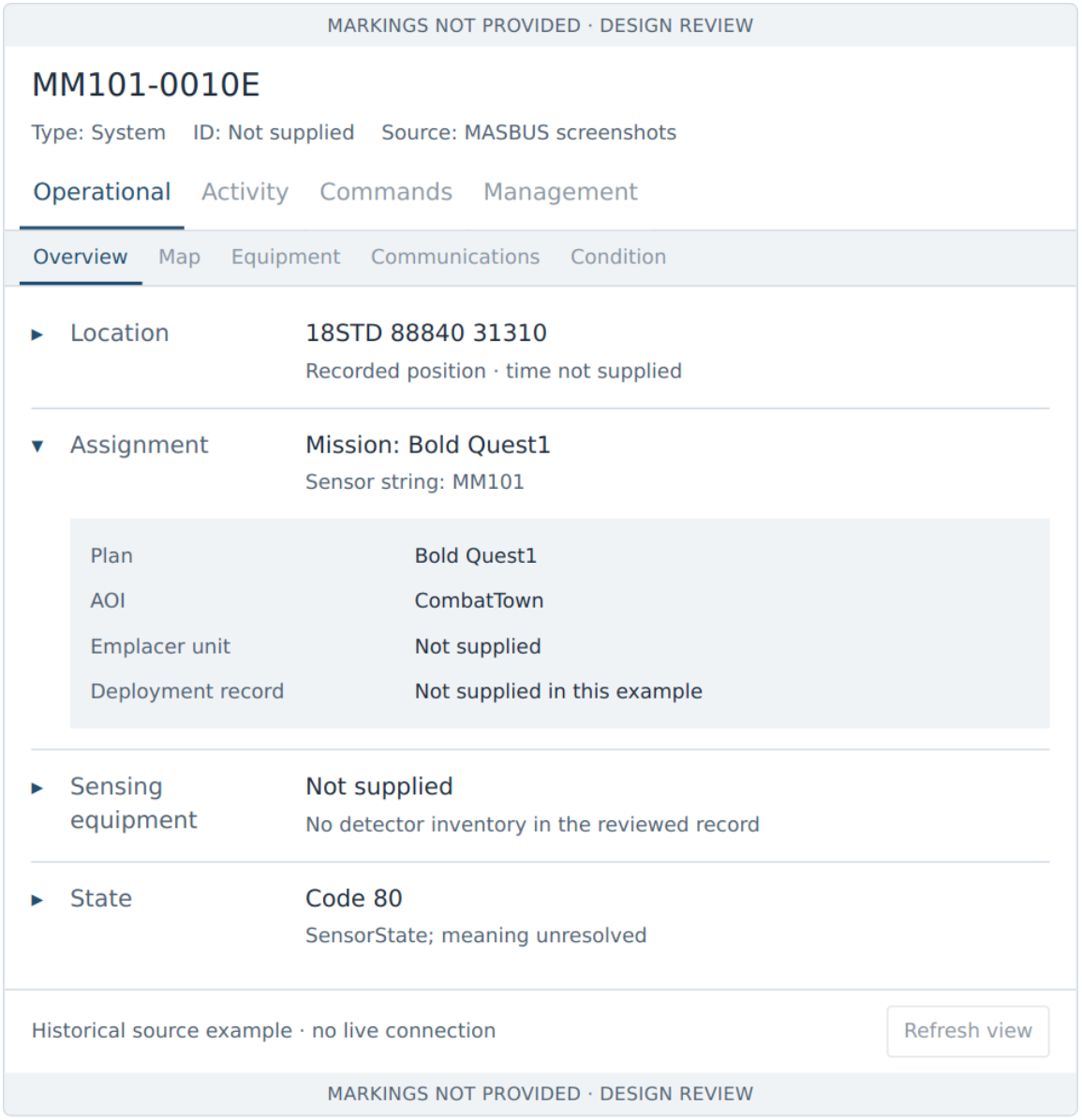
https://www.rfc-editor.org/rfc/rfc7946.html

https://schemas.opengis.net/sensorML/3.0/json/AbstractPhysicalProcess.json

Nine local content, disclosure and layout checks passed. Light/dark and narrow layouts were inspected. These checks do not establish API conformance or live data accuracy. Next review: Assignment, after feedback on this Location proposal.

Assignment: preserve the operational roles

Working question: what mission, sensor grouping or deployment is this system associated with? Assignment is our proposed UI grouping, not a single standard property.



The summary now labels Mission and Sensor string explicitly. Expansion retains Plan and AOI without claiming that matching text means the same thing. These values come from the MASBUS screenshots. The deployment and emplacer gaps remain visible for review; optional blank rows need not become permanent product requirements. [F3, page 40]

TRSS supplies a string designation but no mission in the selected excerpt. OmniSense supplies no corresponding assignment information in the inspected CORE 2 pages; that does not mean the system is unassigned. All examples use the same grouping without transferring values between records. Location remains the proposal from pages 41-43; proceeding did not lock it.

The actual values and their possible carriers

Screenshots establish displayed values and legacy labels. The modern JSON selections below are proposed bindings; no current server payload was inspected.

Fact / evidence	Meaning for this view	Carrier and remaining binding work
Mission: Bold Quest1 MASBUS identifiers and mission classifier [F3]	The mission name recorded for the selected system. It is not its resource ID.	SensorML identifiers[] can carry additional terms, including mission IDs. Select a reviewed definition and read its value. The legacy identifier ends missionName; the repeated classifier does not prove a second assignment. [A1, A2]
Sensor string: MM101 TRSS String Number; MASBUS identifier and characteristic [F1, F3]	A sensor-string designation. The excerpt does not identify a separate string resource or define its membership period.	MASBUS identifier ends sensorString. A modern identifier or defined property can preserve it under a reviewed mapping. The TRSS source path and a shared string definition remain unverified.
Plan: Bold Quest1 MASBUS identifier and characteristic [F3]	A separately labelled plan name. Equal text does not make Plan and Mission synonymous.	Legacy identifier ends planName. Bind its own role. An identifier value does not establish the existence of a retrievable plan document or resource.
AOI: CombatTown MASBUS default details [F3]	A named area of interest in the source presentation. Its geometry and exact operational role are not established.	Serialized carrier not exposed in the inspected full-description screenshots. Do not equate this label with system location, deployment geometry or an observed feature without a defined relation.
Emplacer unit: blank MASBUS explicit role; TRSS agency blank [F1, F3]	Who emplaced the system, when supplied with that role. This differs from owner, operator and generic contact.	SensorML contacts[] supports ResponsibleParty or ContactLink. Bind an agreed role and the relevant party. MASBUS generic contact organisation 2nd GSP does not fill its blank sensorEmplacer role. [A1, A3]
Deployment: none shown All selected examples	An actual employment record, when provided, with its own identity and associations.	CSAPI System links[] relation ogc-rel:deployments leads to related deployments; the nested endpoint is /systems/{sysId}/deployments when that association is provided. [A4]

Exact legacy definition strings visible in MASBUS

urn:ogc:def:identifier:DSOSRI:1.0:missionName
urn:ogc:def:identifier:DSOSRI:1.0:sensorString
urn:ogc:def:identifier:DSOSRI:1.0:planName

These strings are implementation evidence, not certified OGC vocabulary entries. In SensorML 3.0, Term provides definition, label, codeSpace and value; its schema requires label and value. A client that elevates a term into Assignment still needs an agreed meaning or a reviewed source mapping. Schema validity alone does not supply that interpretation. [A2]

A concrete role for Deployment

This is a modeling recommendation for future integration, not a claim that the historical examples already contain Deployment resources.

A sensor string emplaced for a common task and operating period is a defensible Deployment candidate. SensorML explicitly supports deployment-specific identification, contacts, characteristics and per-system description/configuration. [A5] We can use that structure to describe an employment episode while retaining each device's independent System identity. A persistent string name alone is insufficient to establish that episode.

The exact SensorML JSON structure is deployedSystems[] containing system links and optional description/configuration. [A6] CSAPI Deployment validTime is the deployment period; this differs from System validTime, which describes its description's validity. [A4] Do not copy the MASBUS start/end fields into an employment period without confirming their original meaning.

Proposed requirements

ASN-01 - Explicit roles. Keep Mission, Sensor string, Plan, AOI, Emplaced by and Deployment distinct. The UGS summary prioritizes a supplied mission, then string; a supplied deployment can occupy the summary when neither is present. This is a presentation rule, not semantic equivalence.

ASN-02 - Text and links. Readable literals are useful without linked resources. Add navigation only for established identities and associations. Display multiple applicable records as multiple records or a count with expansion; never choose the first silently.

ASN-03 - Time and provenance. Use records applicable to the selected resource/time. Preserve source and role. Consolidate repeated carriers only after confirming they describe the same fact; retain conflicting values with their provenance.

ASN-04 - Read and edit. The overview is readable first. Any edit follows the established metadata, assignment or deployment workflow and permissions. Editing a string label does not by itself move a physical sensor or update a Deployment membership.

ASN-05 - Honest absence. Missing information is not unassigned. Do not derive assignments from radio-network settings, parse them out of a system name, or infer an emplacer from a generic contact. Device-model conformance alone does not define these mission roles.

References and continuity

F1/F3: supplied decks identified on page 40; TRSS slide 11 and MASBUS slides 1-3 were visually re-inspected. A1: DescribedObject.json. A2: commonDefs.json, $defs/Term. A3: ResponsibleParty.json and ContactLink.json. A4: OGC 23-001, Tables 4/10/50 and §11.4.3. A5: SensorML 3.0, §8.9. A6: Deployment.json and DeployedSystem.json.

SensorML 3.0 schemas (A1-A3, A6)

OGC API - Connected Systems Part 1 (A4)

SensorML 3.0 standard (A5)

Ten local content, disclosure and layout checks passed; light/dark and narrow layouts were inspected. Actual bindings and write workflows remain to be validated. Only this working packet was updated. Next: Sensing equipment, keeping configured types, reported transducers and identified attached resources distinct.

What the additional standards contribute

This review corrects an incomplete explanation of their relevance. It does not adopt a universal Assignment concept or certify a cross-standard implementation.

The project lead challenged the apparent absence of useful standards behind Assignment. That concern exposed a gap in our application of the research: the focused wireframe packet had deferred MIMOSA and AAS, then expanded into employment and emplacement without revisiting them. Assignment was an assistant-proposed UI label; proceeding with its review did not make it an approved domain concept.

Established distinction	Verified source	Result for MERLIN
Physical asset and logical equipment role	OPC UA ISA-95 separates Equipment from PhysicalAsset. EquipmentType uses ImplementedBy for the associated asset and AssetAssignment for its history. [R1]	Keep the device identity separate from an enduring role. A reviewed UGS role mapping is still needed; the label MM101 alone does not prove one.
Installation/removal event	MIMOSA OIIE Scenario 11 carries functional location, serialized asset and event time, with optional performing agent and maintenance work order. [R2]	Evaluate this pattern for emplacement history. A functional location is not automatically an AOI, and a maintenance work order is not automatically the MASBUS Plan.
Function, equipment and place as separate perspectives	IDTA 02011-1, based on IEC 81346, separates function, product, location, type and other aspects. Its model defines relationships among aspects. [R3]	Use this as a discipline for separating questions. Do not require every sensor to arrive with five hierarchies, or map a military mission to a function aspect without further definition.
Employment interval and participating systems	SensorML describes deployment purpose, contacts and deployment-specific system information. CSAPI uses Deployments for changeable aggregation. [R4, R5]	Keep Deployment as the OGC resource candidate for an actual employment episode. Its period differs from the instant of installation.

These are complementary contributions, with overlap

MIMOSA helps characterize an action and its participants; ISA-95 helps characterize a role and the asset fulfilling it; the IDTA template separates structural perspectives. Our proposed OGC binding can use those distinctions without requiring three additional runtime systems. A shared word such as assignment, location or plan never establishes equivalence on its own.

Disposition: retain CSAPI/SensorML as the delivery framework. Evaluate these domain definitions where they fit, and specify explicit mappings. No new DISR status, licensing clearance, protocol conformance or production readiness is claimed. Public documentation access alone is not an adoption decision.

Apply the research to the six existing fields

F1/F3 refer to the TRSS and MASBUS source decks already identified on page 40 and inspected for pages 44-46. No new device values are introduced.

Existing fact	Mapping decision and boundary	Presentation consequence
Mission Bold Quest1 [F3]	Retain the legacy missionName term by its reviewed definition. A real mission employment record can be a Deployment candidate; mission text alone cannot identify that record. [R4, R6]	Show Mission explicitly. Add a deployment link only when the association is provided.
Sensor string MM101 [F1, F3]	Preserve the designation. Determine whether the producer means a persistent operational group, a deployment episode, or something else. ISA-95 role/asset separation is useful if enduring positions are actually modeled. [R1]	Show Sensor string. Do not transform its value into a role, platform, Deployment ID or membership period.
Plan Bold Quest1 [F3]	Retain the separately defined planName term. No equality to Mission or MIMOSA maintenance work order is established. A supplied plan document can use a role-qualified SensorML document link. [R2, R6, R10]	Keep Plan visible in disclosure. Provide document navigation only for a supplied link.
AOI CombatTown [F3]	Keep the recorded area label. A Deployment location represents where systems are deployed; an AOI may instead be the area of interest or coverage. These are not automatically the same. [R7]	Show AOI as text. Map or feature navigation needs actual geometry/identity and a defined relation.
Emplacer unit / agency Blank [F1, F3]	A role-qualified contact can preserve a known emplacer. For a timed event, SensorML history[] carries an Event with contacts[] and time; MIMOSA supplies an installation-event pattern to evaluate. [R2, R6, R8, R9]	Display the supplied actor with the relevant event or record. The generic 2nd GSP contact does not fill the blank sensorEmplacer role.
Deployment No record supplied	Use an actual associated Deployment, not a synthesized one from string, plan or mission text. Evaluate meaning, participants and period before choosing this mapping. [R5, R7]	A conditional Deployment row/link becomes useful when the resource exists. Keep absence inspectable without reporting Unassigned.

What this resolves

All six facts now have a disposition. Mission, sensor string and plan remain useful literal facts today. AOI needs a verified relation before mapping. Emplacer has a concrete event-scoped candidate. Deployment has a standard resource path but lacks an instance in these sources. A permanent functional position remains a future modeling decision because none is established by the selected records.

The MASBUS legacy definitions end missionName, sensorString and planName under urn:ogc:def:identifier:DSOSRI:1.0:. They remain historical implementation evidence, not certified vocabulary entries. Do not replace them with new standard names merely because those names sound similar.

The exact OGC paths, and the unresolved parts

Selectors below refer to SensorML 3.0 JSON. They are representation candidates, not recovered TRSS/OmniSense API payloads or a normative MIMOSA/ISA-95-to-OGC crosswalk.

Information	Exact carrier / selector	Binding needed
Recorded names	identifiers[] selected by definition; read value. Term also carries label and optional codeSpace. [R6, R11]	A reviewed term meaning and source/schema version. Valid JSON alone does not identify which term means mission.
Emplacement event	history[]; event.definition, event.time, event.contacts[], event.properties[], event.documentation[]. [R6, R8]	The event kind, owning system, time basis, actor role and any relationship properties. Event requires label and time.
Emplacer in that event	history[n].contacts[m].role and organisationName or individualName. Event contacts use ResponsibleParty. [R8, R9]	An agreed emplacer role URI and the actor identity. This is not the current operator, owner or arbitrary contact.
Legacy emplacer without a timed event	contacts[] on the relevant System or Deployment, selected by a reviewed role. [R6, R9]	Preserve only the supplied assertion. An absent event time cannot be manufactured to create a complete history entry.
Deployment participants	deployedSystems[].system; optional description/configuration; platform is separate. [R7]	Resolved identities and the real relationship. A deployed-system name/description is not a standardized equipment-role identifier.
Deployment period and navigation	Deployment.validTime; System association ogc-rel:deployments and, when provided, /systems/{sysId}/deployments. [R5]	Confirm the interval applies to employment. System description-validity dates are a different time role.
Plan document	documents[].role, documents[].name and documents[].link.href. [R6, R10]	The actual document, relation and link. A planName string is insufficient.

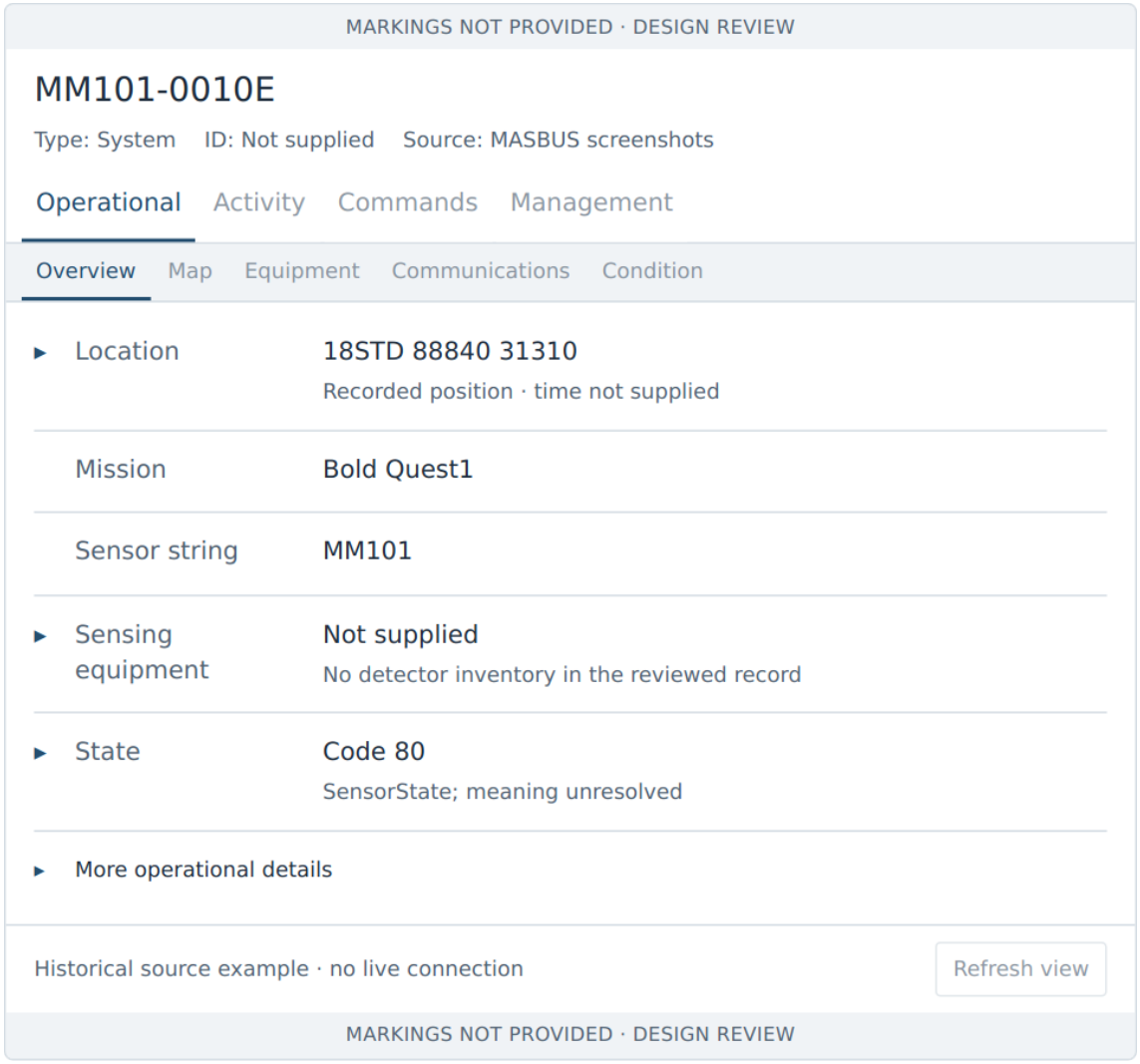
Two implementation details that matter

ISA-95: section 8.2.3.1 explains that assignment Id identifies the counterpart PhysicalAsset or Equipment, depending on the owning object. It is not automatically a separate assignment-record identity. The DataType table names EndTime while the VariableType uses StopTime; any adapter must pin and verify the actual model representation. [R12, R13]

SensorML: a history Event is embedded descriptive information. Rendering it in Activity does not turn it into a first-class CSAPI Observation, create a new event endpoint or prove live subscription support. Our proposed renderer must retain the source kind and reference back to the owning record.

A restrained refinement of the overview

The card uses the actual recorded names while we evaluate the richer relationships. Category tabs and the existing Location proposal remain unchanged.



Mission and Sensor string now have separate labels. More operational details retains Plan, AOI and the explicit source gaps. That disclosure is a presentation mechanism, not a new domain class. It does not assert that Plan and AOI belong to a particular mission resource.

The same ordering applies to the other examples. TRSS shows its string; CORE 2 has no supplied mission/string, so those optional summary rows move to the disclosure as Not supplied. That is an availability rule, not a vendor-specific layout. Their absence does not mean they are inapplicable or that the system is unassigned.

This revision changes presentation only. It adds no inferred asset, role, event, actor, period or Deployment. Actual emplacement events can later appear in Activity, and an explicitly relevant emplacement can be summarized in the overview. A deployment-specific summary must be selected using identity and time, not simply the latest event in an unrelated record.

Decisions ready for the next PBI increment

These are recommendations for review. The previous Assignment screen remains design history; this pass replaces its umbrella label and revises the summary-priority part of ASN-01.

REL-01 - Explicit facts. Use the recorded concept name. Keep literal labels useful even without resource links. Do not require the client to identify every field before it can show any field.

REL-02 - Distinct relationships. Keep physical identity, functional role, membership, installation event and deployment interval distinct. No shared identifier or identical text silently creates an equivalence.

REL-03 - Scoped actor and time. Display who did what and when only from the relevant record. Unknown actor/time remain unknown; role-qualified static metadata remains usable without inventing an event.

REL-04 - Predictable disclosure. Use one stable ordering for supplied mission/string rows. Retain source gaps in More operational details for this review. Final optional-field visibility remains a product decision.

REL-05 - Actions follow meaning. Renaming a label, recording an installation, changing membership and commanding a device are separate operations. Expose each only through the implemented, authorized workflow.

REL-06 - Traceable bindings. For any adopted crosswalk, pin its source version, exact selector, relation direction, subject, time, vocabulary and write behavior. Preserve conflicting assertions with provenance; never silently pick the first.

What is complete, and what is next

The six-field mapping and revised presentation are complete for review. Next, use these distinctions in Sensing equipment: separate configured detector types from identified connected devices, and use the existing CSAPI aggregation rules when actual membership is known. Obtaining one real System description plus related Deployment/history records would validate bindings; it is not a prerequisite to the next design review.

References checked 10 October 2026 (US Eastern)

R1. OPC UA ISA-95 v1.00, EquipmentType, §8.2.4.3

R2. MIMOSA OIIE Scenario 11, Asset Removal/Installation

R3. IDTA 02011-1, v1.0 (Aug 2025), §§1.1, 2.2.3

R4. SensorML 3.0, §§8.2.8, 8.9

R5. CSAPI Part 1, §§10.2, 11.2, 11.4.3

R6-R11. SensorML JSON: DescribedObject; Deployment / DeployedSystem; Event; ResponsibleParty; Document; commonDefs

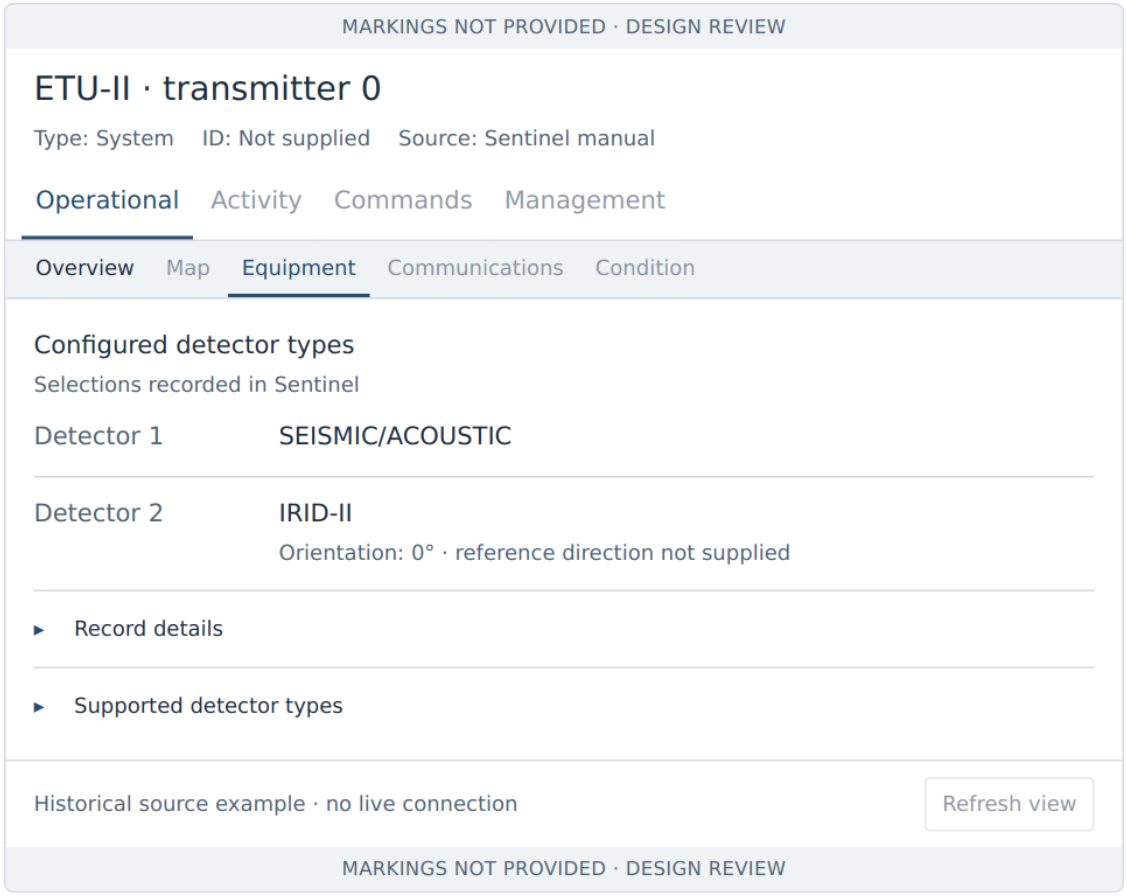
R12. ISA95AssetAssignmentType, §8.2.3.1

R13. ISA95AssetAssignmentDataType, §8.2.2.1

Seven local source-preservation and layout checks passed. This is not live interoperability validation. This working packet is updated; the master research paper, semantics guide, DISR decisions and Power Source workstream are unchanged. Embedded facts/checks preserve this review for continuation.

Equipment: preserve what was selected

The next bounded increment after the relationships review on pages 47-51. Equipment remains a proposed Operational subtab; this pass does not settle every category.



The two detector rows come directly from the TRSS General tab: SEISMIC/ACOUSTIC at Detector 1 and IRID-II at Detector 2. The manual describes choosing the type to be attached. That is enough to present useful configuration information, but it does not identify two physical devices or independently confirm their connection. The IRID-II orientation is preserved as 0 degrees; its reference direction is not established. [F1]

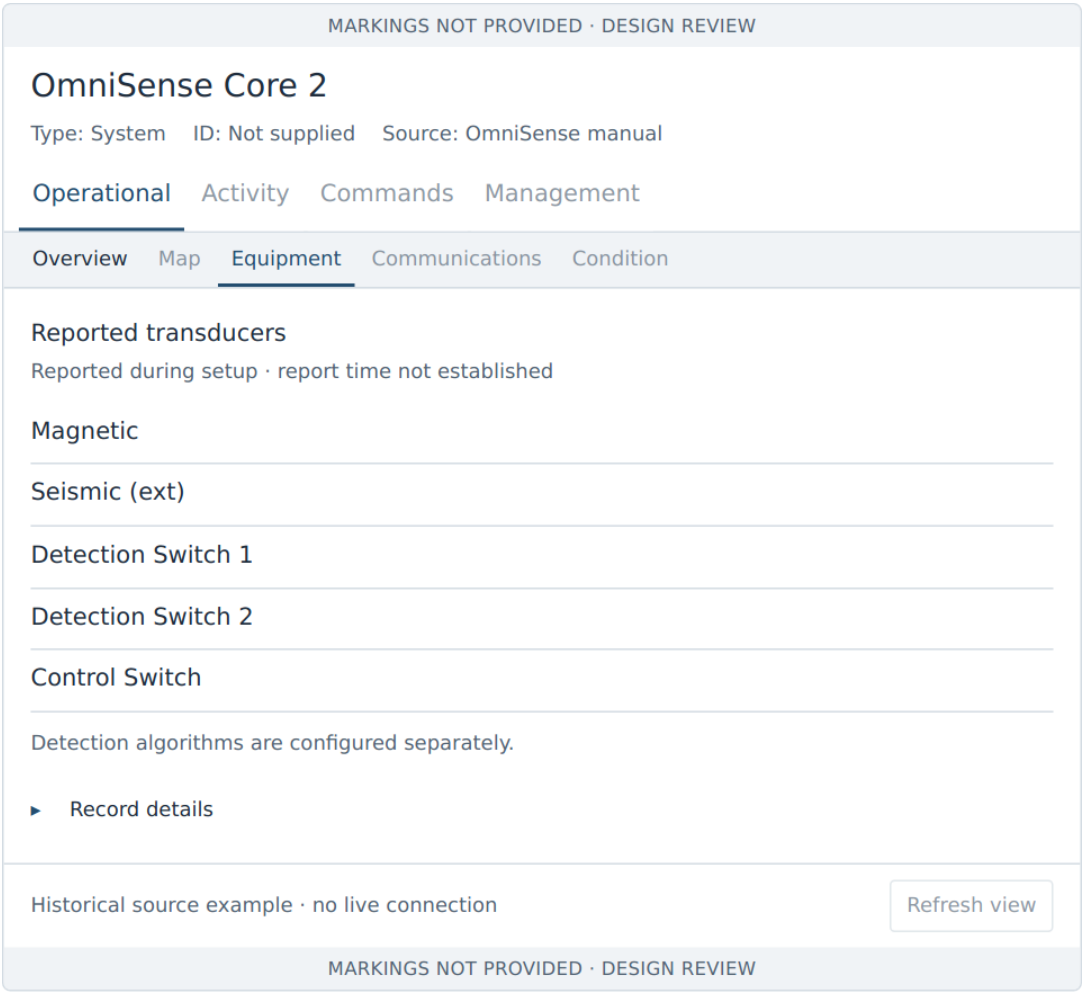
Supported detector types is a separate disclosure. It retains the manual choices for each position, including NONE for Detector 2, and the rule that seismic/acoustic uses Detector 1. Those possibilities never become the selected inventory. These excerpts do not constitute a complete compatibility ruleset. [F1]

This separation applies new research directly: OPC UA Devices distinguishes SupportedTypes from configured instances and applies that pattern to modular devices. It supports a reusable organizing principle; it does not supply an ETU-II vocabulary or prove that Sentinel implements OPC UA. [E1, E2]

The word Equipment on this tab is our user-facing grouping, not a claim that every row is an ISA-95 Equipment object. ISA-95 helps distinguish an enduring logical role from a replaceable physical asset; Detector 1 has not yet been certified as such a role. [E3]

Equipment: preserve what was reported

Same module and subtab, with a different kind of available assertion. The variation follows the data meaning, rather than the vendor name.



OmniSense describes automatic transducer discovery during field setup and transmission of that information to its database. We therefore present the five source labels under Reported transducers. We preserve Detection Switch 1, Detection Switch 2 and Control Switch rather than suppressing them to make the example resemble TRSS. These are five list entries, not evidence of five separately identified physical sensor resources. [F2]

The source provides no connector positions or individual transducer identities. Record details exposes those limits without filling the main view with empty columns. The editor has a Last Updated value, but that does not establish when each transducer was detected. A genuine report timestamp can be shown when its meaning is known. Detection algorithms remain a separate configuration concern. [F2]

The MASBUS example is intentionally sparse: MM101-0010E supplies a system type, but the reviewed record supplies no detector inventory. It shows Equipment information not supplied. It cannot borrow this list, the TRSS selections, or components from the old wireframe. Missing inventory also does not mean no sensors are attached. [F3]

Bind the meaning before choosing the renderer

These are proposed target bindings, not recovered API payloads. Exact paths below refer to SensorML 3.0 JSON and SWE Common 3.0; legacy SensorML 2.0/XML requires its own explicit adapter.

Information we need	Standard structure available	MERLIN decision / remaining binding
Selected detector type at a position	SensorML parameters[] accepts named SWE components. A Category supplies definition, label, value and optional codeSpace / constraint. A DataRecord can group position and type fields. [E5, E7, E8]	Proposed TRSS mapping: select by reviewed definitions and scope to Detector 1 or 2. Do not select fields by label similarity or assume fixed array indexes. The actual definition URIs and payload are still absent.
Settings derived from a base model	SensorML typeOf links the base description; configuration uses Settings. Scalar overrides are configuration.setValues[] entries with ref and value. [E5, E6]	Resolve the targeted base parameter before displaying its effective value. A value override, a parameter definition and a server command are different things. Preserve the original carrier.
Allowed choices and joint constraints	SWE Category.constraint uses AllowedTokens; codeSpace identifies the dictionary. OPC UA DI provides the supported-versus-configured distinction. [E1, E7]	Populate supported choices only from a reviewed model/profile. Per-field token lists alone do not enforce combinations across Detector 1 and Detector 2. Joint restrictions need an explicit validator or service rule.
Reported transducer labels	A SWE DataRecord can define labeled fields; arrays can represent repeated records. [E8]	For an observation-based report, bind the actual DataStream schema and result path, subject and report time. Those paths cannot be recovered from a screenshot. Preserve text until a controlled vocabulary is established.
Identified components and changing attachments	CSAPI distinguishes composition from aggregation. Its subsystem endpoint lists permanent components and may include currently deployed ones; changing attachments use Deployments. SensorML Deployment exposes platform.system and deployedSystems[].system, with per-participant configuration. [E4, E9]	Use actual identities and qualified relationships. No inferred IDs from type labels. An empty subsystem result alone cannot establish an empty interchangeable-sensor inventory. A participant name alone does not define a physical connector.

The small integration contract that makes this reusable

For each supported source/profile version, specify: the source selector and definition; the subject system; whether the assertion is a selection, report, capability or identified relationship; any position/role; the time meaning; and the supported write operation. This is a proposed MERLIN mapping contract, not a newly discovered OGC resource type. Unknown fields remain accessible in the generic standards-oriented view.

If configuration and a report disagree, retain both with their provenance. Do not automatically overwrite one with the other, merge entries by their display names, equate Seismic (ext) with SEISMIC/ACOUSTIC, or relabel a configured component as connected. A policy for choosing a summary may be agreed later; the underlying assertions remain distinct.

Requirements ready for review

This increment completes the source-grounded Equipment presentation proposal. It is not a live integration or a claim that an off-the-shelf cross-vendor UGS profile has been found.

EQ-01 - Compact access. Keep the Overview summary. Its disclosure provides View equipment; the Operational / Equipment subtab shows the fuller list inside the same card. No extra browser page is required.

EQ-02 - Show the assertion accurately. Distinguish configured selections, reported entries, identified attached devices and supported possibilities. Show every supplied entry; do not imply a physical-device count from list length.

EQ-03 - Enrich without redesign. When supplied, add a resolved device link, position/role, identity and relevant time to the existing entry. Join assertions only with verified identity/relationship evidence. A known serial number can remain useful even without a navigable resource.

EQ-04 - Read first; actions are explicit. Editing descriptive configuration, changing an attachment record and sending a device command use their respective implemented and authorized workflows. An editable SensorML parameter does not itself prove that a device can be commanded.

EQ-05 - Keep time and provenance. Separate report time, deployment interval, configuration validity and retrieval time. Refresh reloads available records; requesting a new discovery/status report is a separate supported command. The offline wireframe keeps Refresh visible but disabled.

EQ-06 - Graceful partial support. Unmapped information remains inspectable in the generic view. Absence, explicit NONE, unsupported capability and inaccessible data require different states. Do not fabricate a universal list of detector positions.

EQ-07 - Bounded acceptance. TRSS retains both selections and scoped orientation; CORE 2 retains all five labels; MASBUS gains no borrowed inventory. A later payload-based check must validate selectors, references and time semantics before claiming end-to-end interoperability.

What the research accomplished

We now have a common presentation pattern and named semantic boundaries supported by the device standards. OGC supplies the delivery structures; the reviewed source/profile binding supplies the device-specific meanings. The remaining work is finite: obtain or author one agreed binding for each source and validate it against real records. Another broad standards survey is not required to review this view. Next recommended row: State, using the same evidence discipline.

Evidence and references

F1. TRSS screenshot deck, slides 10-11; Sentinel 1.0.7.2 General tab. F2. SADU deck, slides 6-7; OmniSense manual §§6.1.6.1-2, printed pp. 6-6 to 6-7. F3. MASBUS screenshots, slides 1-3. These are the source artifacts identified on page 40; source images were reinspected for this increment.

E1. OPC UA Devices v1.05.0, §9.2.1, Configurable components

E2. OPC UA Devices v1.05.0, §9.4, Modular devices

E3. OPC UA ISA-95 v1.00, §8.2.1, Role-based equipment

E4. CSAPI Part 1 v1.0, §§10.2, 10.5

E5-E6. SensorML 3.0 JSON: AbstractProcess.json; Settings.json

E7-E8. SWE Common 3.0 JSON: Category.json; DataRecord.json

E9. SensorML 3.0 JSON: Deployment.json; DeployedSystem.json

Checked 11 Oct 2026 UTC. Seven local navigation, source-preservation and layout checks passed; light/dark and narrow layouts visually reviewed. Embedded review facts and checks support continuation. Only this working packet is updated; master research, semantics guide, DISR decisions and Power Source scope remain unchanged.